

# Mechanistic target of rapamycin (mTOR) pathway in Sertoli cells regulates age-dependent changes in sperm DNA methylation

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## Abstract

Over the past several decades, a trend toward delayed childbirth has led to increases in parental age at the time of conception. Sperm epigenome undergoes age-dependent changes increasing risks of adverse conditions in offspring conceived by fathers of advanced age. The mechanism(s) linking paternal age with epigenetic changes in sperm remain unknown. The sperm epigenome is shaped in a compartment protected by the blood-testes barrier (BTB) known to deteriorate with age. Permeability of the BTB is regulated by the balance of two mTOR complexes in Sertoli cells where mTOR complex 1 (mTORC1) promotes the opening of the BTB and mTOR complex 2 (mTORC2) promotes its integrity. We hypothesized that this balance is also responsible for age-dependent changes in the sperm epigenome. To test this hypothesis, we analyzed reproductive outcomes, including sperm DNA methylation in transgenic mice with Sertoli cell-specific suppression of mTORC1 (Raptor KO) or mTORC2 (Rictor KO). mTORC2 suppression accelerated aging of the sperm DNA methylome and resulted in a reproductive phenotype concordant with older age, including decreased testes weight and sperm counts, and increased percent of morphologically abnormal spermatozoa and mitochondrial DNA copy number. Suppression of mTORC1 resulted in the shift of DNA methylome in sperm opposite to the shift associated with physiological aging – sperm DNA methylome rejuvenation, and mild changes in sperm parameters. These results demonstrate for the first time that the balance of mTOR complexes in Sertoli cells regulates the rate of sperm epigenetic aging. Thus, mTOR pathway in Sertoli cells may be used as a novel target of therapeutic interventions to rejuvenate the sperm epigenome in advanced-age fathers.

### eLife assessment

This potentially **important** study addresses the effects of aging on the sperm epigenome and its consequences for reproductive health. The evidence supporting the main claim remains **incomplete**. This study will be of interest to researchers working on aging and reproductive health.

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## Introduction

Over the past several decades, a trend toward delayed childbirth has led to increases in parental age at the time of conception. This delayed parenthood is attributed to secular and socioeconomic factors (Waldenström, 2016 [↗](#)), reproductive technological advancement (Bray et al., 2006 [↗](#)), and higher levels of post-graduate education (Mills et al., 2011 [↗](#)). The suppressive role of overpopulation on reproductive behavior was also hypothesized (Suvorov, 2021 [↗](#)). Emerging evidence suggests that higher male age contributes to poor pregnancy outcomes (Hassan & Killick, 2003 [↗](#)), lower odds of live birth (Horta et al., 2019 [↗](#)), and adverse health of offspring in later life (Montgomery et al., 2004 [↗](#); Puleo et al., 2012 [↗](#); Saha et al., 2009 [↗](#)), including increased susceptibility to early development of cancer (Contreras et al., 2017 [↗](#)), as well as neurodevelopmental, and psychiatric disorders such as schizophrenia (Gratten et al., 2016 [↗](#)) and autism (Reichenberg et al., 2006 [↗](#)). In experiments with mice it was demonstrated that higher male age is associated with reduced life span in offspring (Xie et al., 2018 [↗](#)). Increasing evidence links advanced paternal age with altered offspring phenotype via age-dependent changes in the sperm epigenome (Ashapkin et al., 2022 [↗](#); Jenkins et al., 2014 [↗](#)). Published data indicate that age is a powerful factor affecting DNA methylation and other epigenetic markers in mammalian sperm (Ashapkin et al., 2022 [↗](#)), and age-dependent changes in these epigenetic mechanisms are involved in the regulation of key developmental pathways, including nervous system-related signaling, Wnt, Hippo, mTOR, and Igf1 (Guo et al., 2021 [↗](#); Ma et al., 2020 [↗](#); Pilsner et al., 2021 [↗](#); Suvorov et al., 2020 [↗](#); C. Wu et al., 2020 [↗](#); Xie et al., 2018 [↗](#); Yoshizaki et al., 2021 [↗](#)). Despite this evidence, the mechanistic link between age and epigenetic changes in sperm remains unknown.

Final DNA methylation profiles of spermatozoa are a result of epigenetic events that occur during spermatogenesis (H. Wu et al., 2015 [↗](#)), including meiotic divisions of spermatogonia and spermatocytes associated with passive loss of methylation and *de novo* methylation (Oakes et al., 2007 [↗](#)). Global methylation drop of around 12-13% occurs in preleptotene spermatocytes and methylation is gradually reestablished during leptotene-pachytene stages so that the final DNA methylation patterns are obtained by the end of pachytene spermatocyte stage (Gaysinskaya et al., 2018 [↗](#); Loukinov et al., 2002 [↗](#); Oakes et al., 2007 [↗](#)). A recent rat study identified the transition from elongating spermatids to late spermatids as another stage associated with changes in methylation of more than 5,000 DNA regions (Omri-Charai et al., 2023 [↗](#)). After fertilization, most parental-specific epigenetic marks of gametes undergo reprogramming to establish totipotency in the developing embryo. However, imprinted loci, certain classes of repetitive sequences, and other genomic loci escape this reprogramming event and contribute to a non-Mendelian form of inheritance as demonstrated in humans (Atsem et al., 2016 [↗](#); Denomme et al., 2020 [↗](#)) and rodents (Oluwayiose et al., 2021 [↗](#); Xie et al., 2018 [↗](#)), suggesting that the final sperm epigenome acquired during spermatogenesis is a significant channel for the transfer of inheritable information to the next generation (Lisner & Kimmins, 2023 [↗](#)).

Epigenetic events during spermatogenesis occur in the unique biochemical environment of the apical compartment of seminiferous epithelium, protected from body fluids by the blood-testis barrier (BTB) - the tightest blood-tissue barrier composed of coexisting tight junctions, basal ectoplasmic specializations, gap junctions, and desmosomes in Sertoli cells (**Fig. 1**). The BTB regulates the entry of chemical compounds, including nutrients, hormones, electrolytes, and harmful toxicants, thus creating a unique and highly selective environment for germ cells undergoing epigenetic reprogramming (Mok, Mruk, & Cheng, 2013). In rodents, age-dependent declines in fertility are caused by deterioration of the BTB (Levy et al., 1999; Paul & Robaire, 2013). Recent studies demonstrate that BTB integrity is determined by the balance between activities of the two complexes of the serine/threonine kinase mechanistic target of rapamycin (mTOR) in Sertoli cells, whereby mTOR complex one (mTORC1) promotes disassembly of the BTB and mTOR complex two (mTORC2) promotes its integrity (Li & Cheng, 2016; Mok, Mruk, & Cheng, 2013; Mok, Mruk, Lee, et al., 2013) (**Fig. 1**). mTOR expression is highest in testes out of all tissues in humans (Fagerberg et al., 2014) and mice (Yue et al., 2014), and the mTOR pathway is recognized today as a major mechanism of longevity and aging regulation (Papadopoulos et al., 2019; Weichhart, 2018). Thus, we hypothesized that the balance of mTOR complexes in Sertoli cells may also play a significant role in age-dependent changes in the sperm epigenome. In the current study we test this hypothesis using transgenic mice with manipulated activity of mTOR complexes in Sertoli cells.

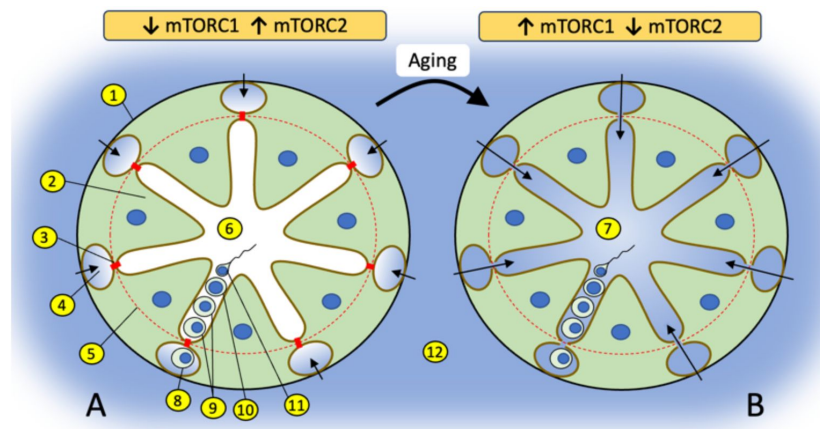
## Material and Methods

### Identification of age-dependent patterns in sperm DNA methylation

To determine patterns of age-dependent DNA methylation change epididymal sperm was collected from C57BL/6 mice at eight different ages: at postnatal days (PND) 56, 80, 107, 118, 190, 225, 265, and 334. Each group consisted of 4 mice except PND 118 and PND 225 groups which included 3 mice each. All animals were purchased from Jackson Laboratories. Sperm DNA methylation was determined using Infinium Mouse Methylation BeadChip (MM285K, Illumina) which covers > 285,000 methylation sites. The methylation of each CpG in the array was estimated based on the intensity of unmethylated and methylated probes. A total of 288,658 probes were identified and were preprocessed using the *sesame* function which included background and dye-bias corrections as well as the removal of SNPs and repeated probes in R. The *minfi* package was used to remove probes that are below background fluorescence levels, adjust the difference in Type I and II probes, and correct for technical variations in the background signals (Niu et al., 2016). The *ComBat* function was used to correct batch effects (Leek et al., 2012). This resulted in a total of 254,410 CpGs for downstream analyses. The results are reported in beta values, which is a measure of methylation levels from 0 to 1. The beta values are converted to M-values by log2 transformation, as this has been reported to be a more significant valid approach for differential methylation analysis and better conforms to the linear model homoscedasticity (Du et al., 2010). To visualize the sperm DNA methylation pattern with age, a time series analysis of all CpGs was then plotted using the Short-Time Series Expression Miner (STEM) (Ernst & Bar-Joseph, 2006).

### Transgenic mice with Sertoli-specific inactivation of mTORC1 or mTORC2

To inactivate mTORC1, we used a Raptor (component of mTORC1) knockout model. To inactivate mTORC2, we used a Rictor (component of mTORC2) knockout model. To achieve Sertoli-specific knockouts, mice with floxed Rictor or Raptor (Rictor<sup>tm1.1Klg/SjmJ</sup> and B6.Cg- Raptor<sup>tm1.1Dmsa/J</sup> respectively) were crossed with mice with Cre recombinase controlled by the *Amh* Sertoli-specific promoter (129S.FVB-Tg(*Amh-cre*)8815Reb/J). All mice were purchased from Jackson Laboratories (Stock #: 020649, 013188 and 007915 respectively). In short, to generate Raptor Sertoli-specific KO



**Fig. 1.**

**Changes in the balance of mTOR complexes and aging affect biochemical conditions of spermatogenesis in the apical compartment of seminiferous epithelium.**

**(A)** Seminiferous tubule with tight BTB where interstitial environment (shown in blue) penetrates only to basal compartment of seminiferous epithelium as show by color and arrows. The tight state of the BTB is promoted by higher activity of mTORC2 over mTORC1 in Sertoli cells. **(B)** Seminiferous tubule with leaky BTB where interstitial environment penetrates to the apical compartment of seminiferous epithelium. BTB disassembly is promoted by higher activity of mTORC1 over mTORC2. (1) basal membrane of a seminiferous tubule; (2) Sertoli cell; (3) contact between Sertoli cells enforced by gap junction, tight junction and endoplasmic specialization (BTB); (4) basal compartment of the seminiferous epithelium; (5) the dotted red line illustrates structural separation of compartments by the BTB; (6) apical compartment of the seminiferous epithelium; (7) apical compartment in aged organism with biochemically “noisy” environment due to the leaky BTB; (8) spermatogonia; (9) spermatocytes; (10) spermatid; (11) spermatozoa; (12) interstitial environment.

mice, male B6.Cg-Rptor<sup>tm1.1Dmsa/J</sup> mice were paired with female 129S.FVB-Tg(Amh-cre)8815Reb/J mice. In the next round of breeding F1 females were paired with the same B6.Cg-Rptor<sup>tm1.1Dmsa/J</sup> males. In F2 offspring around 25% had genotype Rptor<sup>flox/flox</sup>, and 25% had genotype Rptor<sup>flox/flox</sup>, Amh<sup>cre/+</sup>. Animals with these two different genotypes were paired together to produce F3 mice, which were used for sperm collection and reproductive phenotype assessment. F3 male mice were represented by 2 genotypes: Sertoli-specific KO of Raptor (suppressed mTORC1), and wildtype (WT) animals used as controls. The same breeding scheme was used to produce F3 male mice with Sertoli-specific KO of Rictor (suppressed mTORC2) and their corresponding WT siblings. All animals used in breeding were 9-10 weeks old. Breeding was done by pairing one male and two female breeders overnight.

Females were inspected for vaginal plugs at 9 am every morning and presence of the plug was considered pregnancy day 1. Pregnant females were housed individually. At weaning on postnatal day 21 all animals were identified via ear-punching, and ear tissue samples obtained from ear-punching were used for genotyping. Genotyping was done using RT-qPCR as recommended by the Jackson Laboratories. All animals were housed in a temperature (23 ± 2°C), and humidity (40 ± 10 %) controlled environment, with a 12-h light/dark cycle, and food and water available ad libitum. F3 male mice were euthanized at 4 time-points: 8, 12, 22 and 56 weeks of age (n ≥ 5 per genotype/timepoint). At each euthanasia testes were collected, weighed, and fixed in modified Davidson's solution and caudal epididymal sperm was collected via swim-up procedure (see below, Sperm collection and DNA extraction). All procedures followed the guidelines of the National Institutes of Health Guide for the Care and Use of Laboratory Animals and the approval for this study was received from the Institutional Animal Care and Use Committee at University of Massachusetts, Amherst.

## Immunohistochemistry

Testes samples collected from 22-week-old mice were fixed in modified Davidson's fluid, dehydrated through a series of alcohols and xylene, and embedded with Paraplast X-tra paraffin (Leica Biosystems) using the Excelsior ES Tissue Processor (Thermo Fisher). Five-micrometer sections were cut on a Microm HM 355S microtome (Fisher) and mounted on Colorfrost Plus slides (Fisher). Immunohistochemistry (IHC) was performed on a DakoCytomation autostainer, using the Envision HRP Detection system (Dako, Carpinteria, CA). Sections were deparaffinized in xylene, rehydrated in graded ethanols, and rinsed in Tris-phosphate-buffered saline (TBS).

Heat induced antigen retrieval was performed in a microwave for 20 minutes at 98°C in 0.01 M citrate buffer (pH6) for Rictor antibody and Tris-EDTA (pH9) for Raptor antibody. After cooling for 20 min, sections were rinsed in TBS and subjected to the primary rabbit polyclonal anti-Rictor (1:100, Abcam, ab70374), rabbit monoclonal anti-Raptor [EP539Y] (1:100, Abcam, ab40768) for 60 and 30 minutes respectively. After subsequent washes in TBS, slides were incubated with secondary antibody (DAKO Envision Flex+ anti-rabbit polymer, K800921-2) for 20 minutes. Immunoreactivity was visualized by incubation with chromogen diaminobenzidine (DAB) (DAKO, K3468) for 10 min. Tissue sections were counterstained with Mayer's hematoxylin (Poly Scientific R&D Corp, S216), dehydrated through graded ethanols and xylene, and cover-slipped. Sections were viewed through a Zeiss Axio Observer Z1 inverted light microscope with ZEN imaging software. A minimum of 20 images per slide were captured at 88000 dpi using the AxioCam 506 color digital camera and analyzed for differences in antibody staining in seminiferous tubules at different stages.

## BTB permeability analysis

The permeability of the BTB was assessed in all WT and KO mice euthanized at 22 weeks of age using a biotin tracer assay as described elsewhere (25). In short, animals were anesthetized with isoflurane, their testes were exposed, a small opening was created in the tunica albuginea of the right testes, and 50 µl of 10 mg/ml EZ-Link Sulfo-NHS-LC-Biotin (Life Technologies) in PBS

containing 1 mM  $\text{CaCl}_2$  was injected into the interstitium. After 30 min, mice were euthanized, and their testes were immediately removed and fixed in modified Davidson's solution. Testis were embedded in paraffin and 5- $\mu\text{m}$  sections. Deparaffinized slides were incubated with 5  $\mu\text{g}/\text{ml}$  Streptavidin Alexa Fluor 405 conjugate in PBS at 25°C for 1 hour, mounted using Diamond Antifade Mountant (P36965), and incubated for 24 hours at room temperature in dark room. The imaging was performed using A1R-SIME confocal microscope (Nikon) at the light microscopy facility at the Institute of Applied Life Sciences, University of Massachusetts Amherst.

## Sperm collection and DNA extraction

At each euthanasia, the right and left cauda epididymides were incised, cut three times, and incubated at 37°C for 30 minutes in 1 ml of sperm wash buffer (Cat. # ART1006, Origio, Denmark). After incubation the epididymides were removed, the tube was vortexed for 10 sec and 10  $\mu\text{l}$  of each sample was smeared evenly across the whole surface of the glass slide for microscopic analysis. The remaining samples were used for DNA extraction following the rapid method developed in J.R. Pilsner's laboratory (43). In short, sperm samples were subjected to a one-step density gradient centrifugation over 40% Pureception buffer (CooperSurgical Cat # ART-2100) at 600g for 30 min in order to remove possible somatic contamination. Sperm cells were homogenized with 0.2 mm steel beads at room temperature in a mixture containing guanidine thiocyanate lysis buffer and 50 mM tris(2-carboxyethyl)phosphine (TCEP). The lysates were column purified using QiaAMP DNA mini-Kit (Qiagen, Cat # 56304) and DNA quality of the eluate was determined using Nanodrop 2000 Spectrophotometer (#E112352; Thermo Scientific, Somerset, NJ). The DNA samples were stored at -80°C.

## Sperm quality analysis

Sperm smears were fixed by immersing in 3% glutaraldehyde in PBS (PH=7.2) for 30 minutes, washed briefly in PBS (PH=7.2) and allowed to dry at room temperature. Smears were stained by 5% aniline blue in 4% acetic acid (PH=3.5) for 5 minutes, washed briefly in PBS (PH=7.2) and allowed to dry at room temperature. Slides were covered with a cover slip using mounting solution incubated at room temperature for 30 minutes and examined using an inverted light microscope at 40 X magnification. The sperm quantity was assessed by counting spermatozoa in 10 random windows on each slide. For sperm morphology analysis, 200 spermatozoa were counted, and number of abnormal spermatozoa was recorded.

## Mitochondrial DNA Copy Number

Sperm DNA was used to measure mitochondrial copy number via real-time quantitative PCR using a published protocol (44). The primers used to calculate relative mitochondrial copy number include a primer pair for the mitochondrial genome (5'-CTCCGTGCTACCTAAACACCTTATC-3' and 5'-GACCTAAGAAGATTGTGAAGTAGATGATG-3') and another primer pair for nuclear genome for a single copy Apob gene (5'-CACGTGGGCTCCAGCATT-3' and 5'-TCACCAGTCATTCTGCCTTTG-3'). Triplicate 5- $\mu\text{l}$  real-time PCR reactions, each containing iTaq Universal SYBR Green Supermix (Cat 172-5124, BioRad), primers, and cDNA template were loaded onto a 384-well plate and run through 40 cycles on a CFX384 real time cycler (Bio-Rad Laboratories, Inc). The data were analyzed using the manufacturer's CFX manager software, version 3.1. Relative quantification was determined using the  $\Delta\Delta\text{C}_q$  method.

## Library Preparation and Reduced Representation Bisulfite Sequencing (RRBS)

Sperm DNA methylation was analyzed at 8 and 22 weeks of age, with four randomly selected individual samples per age and genotype (Rictor KO, Rictor WT, Raptor KO, Raptor WT).



Bisulfite conversion was performed on 100 ng of genomic DNA using the EpiTect Fast DNA Bisulfite kit from Qiagen (Cat.# 59824) and DNA libraries were constructed using the NUGEN Ovation RRBS Methyl-seq System (Cat.# 0353) according to the manufacturer's instructions. Libraries were sequenced on Illumina HiSeq 4000 at the Deep Sequencing Core Facility of the University of Massachusetts Medical School (Shrewsbury, MA) with an average of 22.7 million single-end reads (50-bp) per sample.

## RRBS data analysis

Raw reads were trimmed using TrimGalore (v 0.6.6) and a NuGEN-specific adaptor trimming scripts available from GitHub (nugentechnologies/NuMetRRBS). Trimmed reads were mapped using Bismark-Bowtie2 with no mismatch allowed. Methylation counts were called using Bismark extract. Differentially methylated regions were identified using the MethylKit (v 1.24.0) pipeline (45). In brief, the genome was tiled into sliding windows and a weighted methylation level was calculated for each window. To minimize error in base calling, we filtered out bases with less than 10x coverage and read counts more than 99.9<sup>th</sup> percentile and coverage values were normalized using default settings. We used a logistic regression model for *p*-value calculation subsequently adjusted for multiple comparison (FDR) using the SLIM method for final DMR identification. Individual DMRs were identified for a 100 bp sliding window with a minimum of 1 CpG in at least 3 samples out of 4 in an age/genotype group. DMRs with methylation difference at FDR < 0.05 were used to compare different age/genotype groups. To compare the overlap between DMRs induced by age and DMRs induced by KO we used Fisher's exact test. For overlapping DMRs we used Pearson correlation to establish concordance of DNA methylation change induced by aging and genetic manipulations. Genomic ranges were compared using the FindOverlapOfPeaks function available in the chiPeakAnno package (46). The total number of methylation regions used as a background was determined as a list of 100 bp tiles containing CpGs obtained by intersecting lists from two respective comparisons (e.g. WT8 vs WT22 and WT22 vs KO22 in mTORC1 experiment and WT8 vs WT22 and WT8 vs KO8 in mTORC2 experiment). Spatial genomic annotation was conducted using annotatr package (v1.24.0) (47) and annotated to the genomic features of the Ensembl genome. Genomic features were compared using the GenomicRanges package (v1.50.2) (48). Each DMR was assigned to the closest gene (<5 kb upstream transcription start site, promoter, 5'UTR, exon, or intron) or intergenic region, and graphs were plotted using ggplot2 package (v3.4.0). We used Metascape (29) to analyze biological categories associated with genes annotated to DMRs. For global methylation changes DMRs were analyzed for a 1000 bp sliding window with at least 3 CpGs and methylation difference > 10% at FDR < 0.05. The original RRBS data files (fastq) are available via Dryad (28).

## Statistical analysis

For all sperm parameters differences between KO and corresponding WT age groups were identified using T-test and all *p*-values were corrected using Benjamini-Hochberg procedure to account for multiple comparison. We considered *q* (FDR corrected *p*)  $\leq$  0.05 as statistically significant difference. Statistical approaches for bioinformatic analyses are described in corresponding sections above.

# Results

## Patterns of age-dependent changes in sperm DNA methylation

Recent comprehensive review of age-dependent changes in sperm DNA methylation in humans and animal models concluded that existing evidence is contradictory and it does not provide clear understanding if most of DNA methylation changes in sperm are linearly associated with age or if these relations are more complex (Ashapkin et al., 2022 [49](#)). Thus, we characterized age-dependent changes in methylation of 254,410 CpGs sites over eight time-point covering the period from 56 to

334 PND using BeadChip methylation arrays and STEM time series analysis which selects temporal profiles independent of the data and identifies significance of enrichment of clusters of similar profiles (Ernst & Bar-Joseph, 2006 [↗](#)). STEM analysis demonstrated that only two clusters were enriched significantly representing linear increase and linear decrease in methylation over the entire analyzed age-period (**Fig. 2** [↗](#)). We repeated this analysis for the top 10% CpGs with highest variance across all ages. The results were the same as analysis done for all CpGs – only clusters of linear age-dependent methylation increase, and decrease were enriched significantly (data not shown). These results suggest that within the studied period (56-334 PND), methylation of most DNA regions undergoing age-dependent changes in methylation in sperm is associated with age linearly or semi-linearly. Thus, within the studied period, comparison of any two age- groups, distant enough to detect age-dependent change, may be used to characterize sperm methylome aging as well as sperm methylome aging acceleration/deceleration.

## Approach to manipulate the balance of mTOR complexes

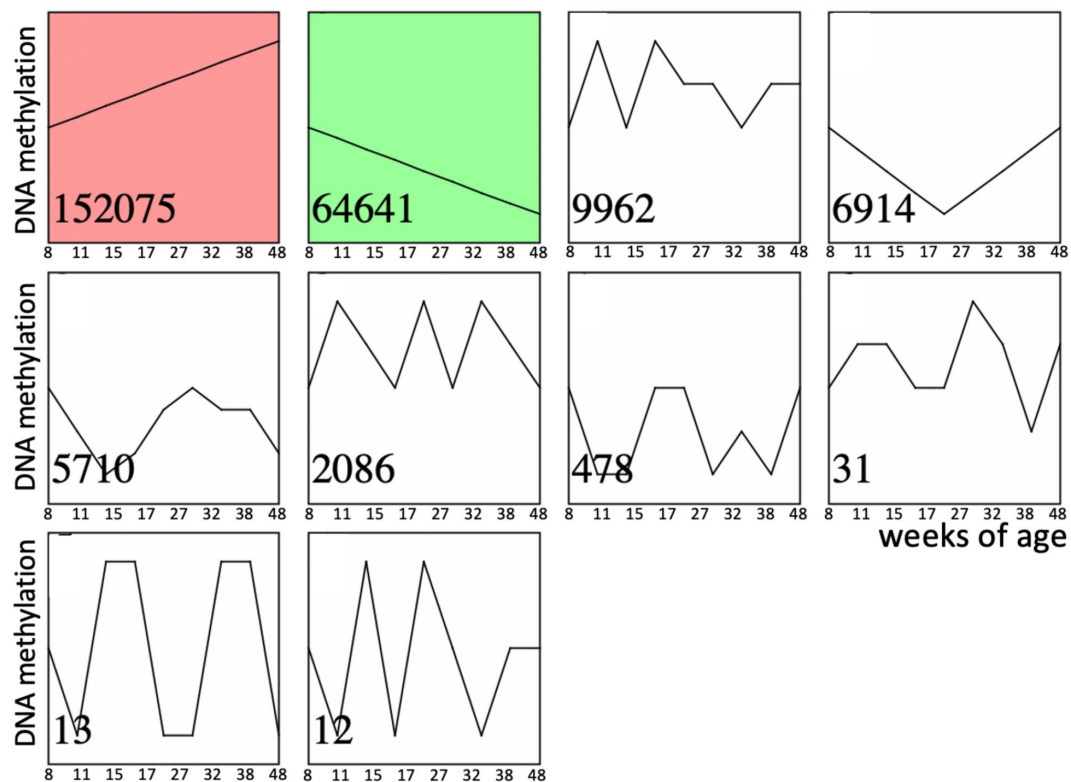
To study if shifting the balance of mTOR complexes activity in either direction in Sertoli cells will induce sperm DNA methylation and other reproductive changes concordant with accelerated or decelerated aging, we generated two transgenic mouse models with manipulated mTOR complexes activity. We used the Cre-Lox approach to knockout (KO) critical components of mTORC1 or mTORC2 – Raptor and Rictor, respectively. In both experiments, Cre recombinase was controlled by the anti-Mullerian hormone (Amh) promoter to insure KO of Raptor or Rictor in Sertoli cells only (**Fig. 3** [↗](#)). We used immunohistochemistry with antibodies for Raptor and Rictor to test efficiency of our genetic manipulations. Indeed, staining of Sertoli cells with Raptor and Rictor antibodies was abolished in corresponding KO animals (**Fig. S1** [↗](#)). BTB permeability was measured in 22-week old KO mice and their wildtype (WT) siblings using a biotin tracer assay, where biotin is injected into the testis interstitium *in-vivo*, and its diffusion into the apical compartment of seminiferous epithelium is visualized in histological sections by streptavidin conjugated with a florescent dye (Meng et al., 2005 [↗](#)). As predicted, in mice with suppressed mTORC2 the BTB was loose enough to allow biotin to migrate to the lumen of seminiferous tubules (**Fig. 4** [↗](#)).

## Age dependent changes in sperm methylome affect methylation of developmental genes

We used reduced representation bisulfite sequencing (RRBS) to analyze DNA-methylation in sperm to test if manipulating mTOR complexes balance may accelerate or decelerate epigenetic aging of sperm. For this analysis we first identified regions of DNA that undergo significant changes in methylation in wildtype animals. To do so, we compared profiles of DNA methylation between 8 and 22-weeks old WT mice in each of the 2 experiments (**Fig. 3** [↗](#)). These ages correspond to young pubertal and mature adult stages in humans (Bell, 2018 [↗](#); Flurkey et al., 2007 [↗](#)). The first age (8 weeks) was selected to avoid sperm from the first wave of spermatogenesis be included in the analysis to ensure that observed changes are truly aging dependent rather than associated with different stages of reproductive system maturation.

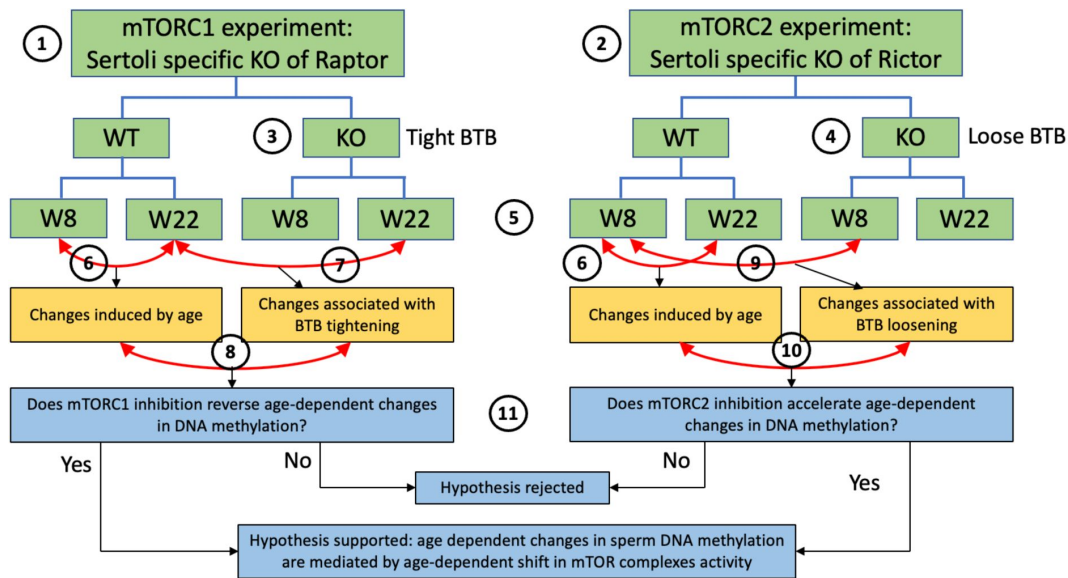
C57BL/6 mice first have fertile sperm in cauda epididymis at 37 days of age (Mochida et al., 2019 [↗](#)), and young C57BL/6 mice ejaculate spontaneously around 3 times per 5 days (Huber et al., 1980 [↗](#)). Thus, no sperm from the first wave of spermatogenesis may survive in their cauda epididymides to the age of 8 weeks. Methylation changes were analyzed for 100 bp windows containing a minimum of 1 CpG. Average number of CpGs per region was  $10.18 \pm 0.08$  and  $10.32 \pm 0.09$  (mean  $\pm$  SE) in mTORC1 and mTORC2 experiments respectively (**Fig. S2** [↗](#)).





**Fig. 2.**

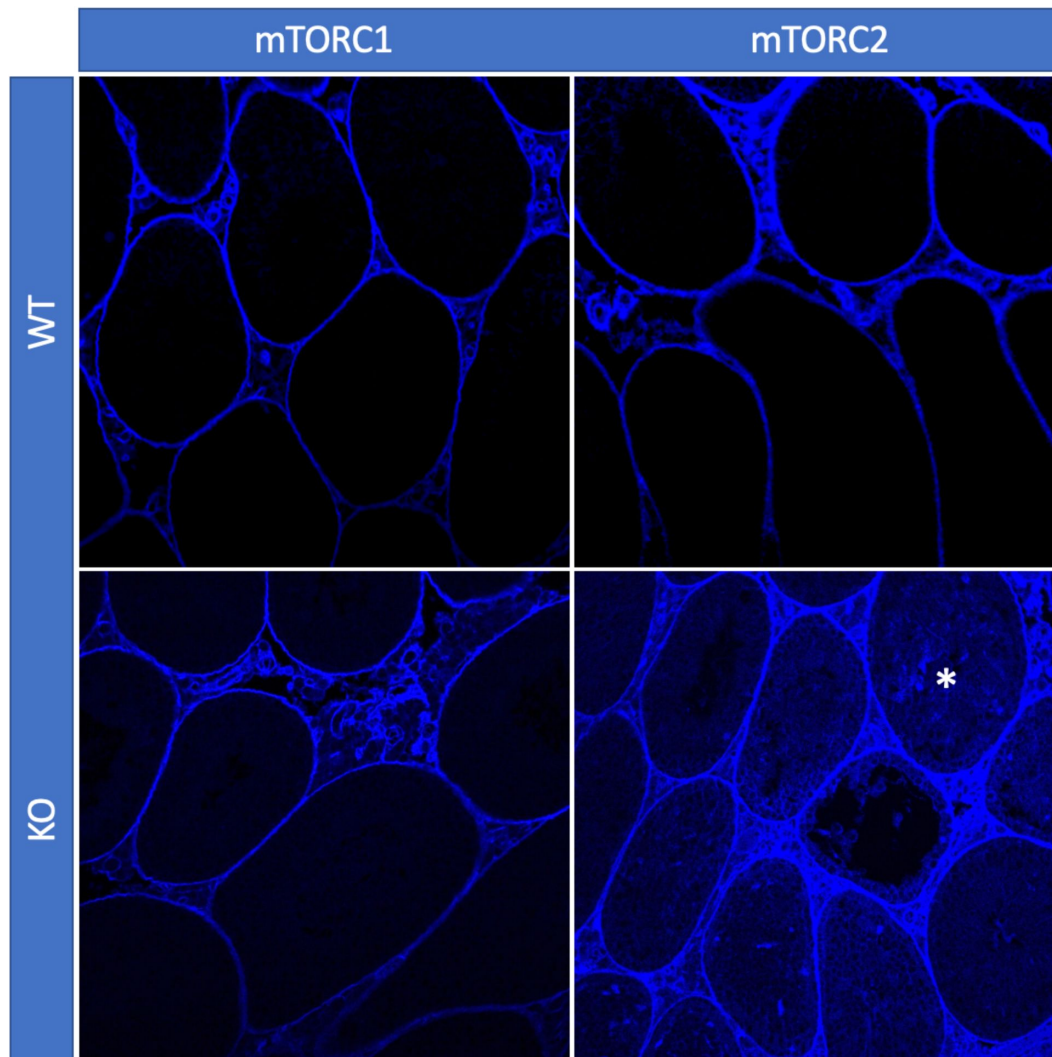
Time series of age-dependent DNA methylation change in sperm. Significantly enriched time-series are shown in color. Numbers in the left bottom corner of each time series indicate the number of corresponding CpGs out of total 254,410 CpGs analyzed.



**Fig. 3.**

### Scheme of experimental design for DNA methylation analysis.

This study utilized experiments with two transgenic mouse models: (1) one with mTORC1 suppression in Sertoli cells due to the cell-specific KO of Raptor, and (2) another with mTORC2 suppression in Sertoli cells due to the cell-specific KO of Rictor. Suppression of mTORC1 results in tightening of the BTB (3) and suppression of mTORC2 results in loosening of the BTB (4). DNA methylation changes were analyzed in sperm of each genotype on postnatal weeks 8 and 22 (5). In each experiment, we first identified physiological age-dependent changes in sperm DNA methylation by comparing epigenomes of WT 8-week old mice and WT 22-week old mice (6). We further used both experiments to test the hypothesis that age-dependent changes in the sperm epigenome are associated with the age-dependent increase in permeability of the BTB. Specifically, to test this hypothesis using the mTORC1 experiment, we compared physiological age-dependent changes (6) with changes induced by KO (tighter BTB) in older mice (7). Our hypothesis predicts that mTORC1 suppression in older mice will affect age-dependent DMRs in the direction opposite to the one induced by age (8). Similarly, to test our hypothesis using the mTORC2 experiment, we compared physiological age-dependent changes (6) with changes induced by KO (loose BTB) in younger mice (9). Our hypothesis predicts that mTORC2 suppression in young mice due to KO will produce similar effects on age-dependent DMRs as age itself (10). Outcomes of both experiments were used to support or reject our hypothesis (11).



**Fig. 4.**

**BTB permeability in 22-week old mice of different genotypes, representative images.**

In mice with inactivated mTORC2 (Rictor KO), biotin tracer (blue) penetrated to the apical compartment of the seminiferous epithelium (asterisk) while in all other genotypes it did not cross the BTB.

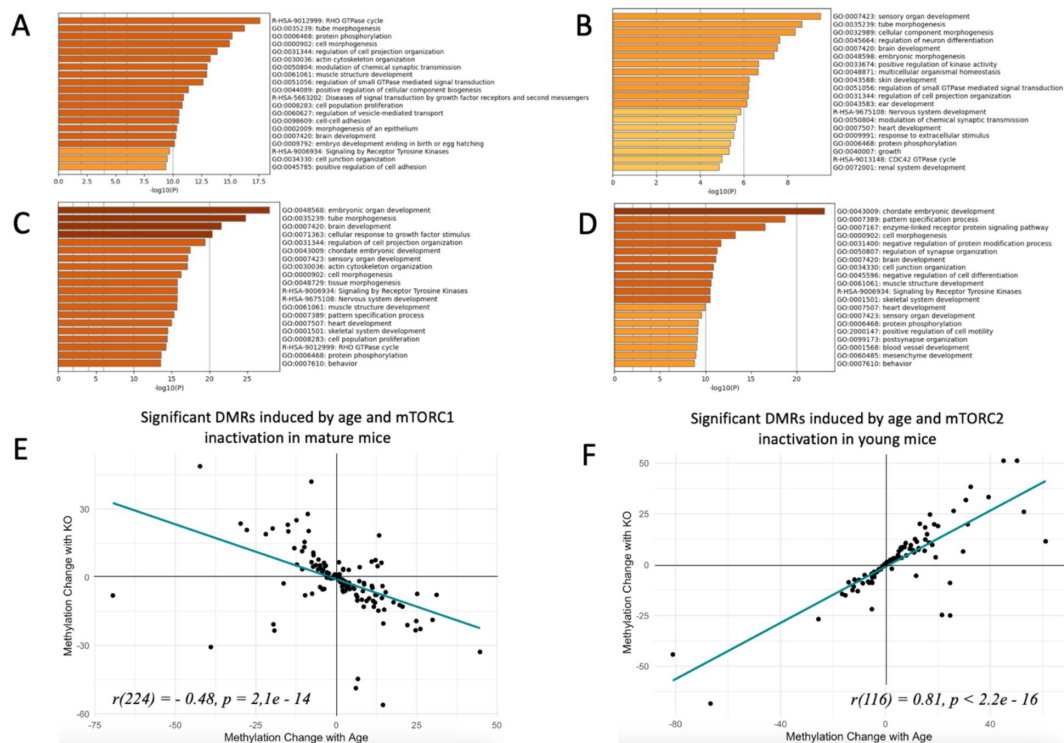
Significant DMRs were defined as any 100 bp window with methylation difference at  $FDR < 0.05$ . Using this approach, we identified 1,731 age-dependent DMRs in the mTORC1 experiment and 797 DMRs in the mTORC2 experiment, with 79 common DMRs identified in both experiments. In both experiments age-dependent changes were dominated by an age-dependent increase in methylation (**Fig. S3A,B**). We assigned each DMR to a gene with overlapping gene body or to a closest gene with transcription start site within 5000 bp downstream of the DMR (Amir et al., 2023) and used Metascape (Zhou et al., 2019) to analyze biological categories associated with differentially methylated regions. Furthermore, we determined genomic elements associated with significant DMRs (**Fig. S4A,B**). This analysis demonstrated that in both experiments, lists of genes associated with significant age-dependent DMRs are enriched with developmental categories (**Fig. 5A,B**), indicative of the possibility that age-dependent changes in sperm DNA methylation may be causative of the transfer of different developmental epigenetic information to the next generation. These findings are concordant with previous research (Pilsner et al., 2021; Suvorov et al., 2020; Xie et al., 2018). Specifically, common highly enriched categories across both experiments included: tube morphogenesis, brain development, cell morphogenesis, and other categories relevant to growth, embryo development, development of nervous system, muscles, and others. Enriched molecular mechanisms were dominated by small GTPase mediated signal transduction and regulation of kinase activity (protein phosphorylation, signaling by receptor tyrosine kinase, and other). The complete results of enrichment analysis can be found at Dryad (Amir et al., 2023).

## Decreased activity of mTORC1 “rejuvenates” sperm methylome

At the next step, we analyzed if Sertoli-specific inactivation of mTORC1 affects methylation of age-dependent DMRs. Given that age-dependent changes in the sperm methylome are associated with age-dependent increase in the BTB permeability, we assumed that decreased permeability of the BTB due to genetic manipulation may result in sperm epigenome “rejuvenation”. Thus, our hypothesis predicts that DNA regions that undergo methylation changes in wildtype animals with age will undergo the opposite change in older animals with Sertoli-specific inactivation of mTORC1 as compared to WT mice of the same age (**Fig. 3**). In other words, DMRs that change with age will be returned to their younger state by mTORC1 inactivation. We identified 2,738 significant DMRs induced by mTORC1 inactivation in mature mice (Amir et al., 2023) and compared them with age-dependent DMRs in WT animals. Most of DNA regions induced by genetic manipulation of Raptor undergo hypomethylation, - the direction of change opposite to the one induced by age (**Fig. S3C**). The distribution of genic elements associated with DMRs was similar in relation to age and genetic manipulation, suggesting that both factors affect the similar sets of elements (**Fig. S4C**). Next, we analyzed an overlap between significant age- dependent and KO-dependent DMRs in the list of all methylation regions identified in both comparisons. According to this analysis the two lists were highly overlapping with  $p < 0.00001$ , indicative that age and mTORC1 inactivation in Sertoli cells affect the similar sets of DMRs in sperm (**Fig. S5A**). Finally, we analyzed the direction of change of overlapping DMRs (**Fig. 5E**). We observed a strong negative correlation between methylation changes induced by age and changes induced by mTORC1 inactivation ( $r = -0.48$ ,  $p = 2.1e-14$ ), suggesting that in mature animals, mTORC1 inactivation in Sertoli cells returns age-dependent DMRs to their younger state. The correlation was even higher for DMRs that undergo 10% or higher change in relation to age or genetic manipulation ( $r = -0.68$ ,  $p = 2.3e-4$ , **Fig. S5C**).

## Decreased activity of mTORC2 accelerates epigenetic aging of sperm

Similarly, we assumed that if age-dependent changes in the sperm methylome are associated with age-dependent increase in the BTB permeability, then increased permeability of BTB due to Sertoli-specific inactivation of mTORC2 may be associated with accelerated aging of the sperm epigenome. Thus, our hypotheses predicts that DNA regions that undergo methylation changes in wildtype



**Fig. 5.**

## Changes in sperm DNA methylation induced by age and manipulation of mTOR pathway in Sertoli cells.

(A, B, C, D) Top biological categories enriched with genes associated with significant DMRs: (A) age-dependent DMRs in wildtype animals in the mTORC1 experiment; (B) age-dependent DMRs in wildtype animals in the mTORC2 experiment; (C) DMRs induced by mTORC1 inactivation in mature animals; (D) DMRs induced by mTORC2 inactivation in young animals. (E and F) Changes in methylation of significant DMRs induced by age and mTORC1 inactivation in mature mice (E) and by age and mTORC2 inactivation in young mice (F).

animals with age will undergo similar changes in young animals with Sertoli-specific inactivation of mTORC2 as compared with same age young WT mice (**Fig. 3**). We identified 1,632 significant DMRs induced by decreased mTORC2 activity in Sertoli cells of young mice (Amir et al., 2023). The majority of the DNA regions with methylation changes induced by genetic manipulation of Rictor undergo hypermethylation, - the direction of change concordant with accelerated sperm aging (**Fig. S3D**). The distribution of genic elements associated with DMRs was similar in relation to age and genetic manipulation, suggesting that both factors affect similar sets of elements (**Fig. S4D**). Further, we conducted an overlap analysis between significant age-dependent and KO-dependent DMRs in the list of all methylation regions identified in both comparisons. According to this analysis the two lists were highly overlapping with  $p < 0.00001$ , indicative that age and mTORC2 inactivation in Sertoli cells affect the similar sets of DMRs in sperm (**Fig. S5B**). Finally, we analyzed the direction of change of overlapping DMRs (**Fig. 5F**). We observed a strong positive correlation between methylation changes induced by age and changes induced by mTORC2 inactivation ( $r = 0.81$ ,  $p < 2.2e-16$ ), suggesting that in young animals, mTORC2 inactivation in Sertoli cells accelerates epigenetic aging of sperm. The correlation was even higher for DMRs that undergo 10% or higher change in relation to age or genetic manipulation ( $r = 0.83$ ,  $p < 2.6e-8$ , **Fig. S5D**).

## Manipulations of mTOR complexes affect methylation of developmental genes

Given our data demonstrate that DNA methylation changes induced by suppression of mTOR complexes affect age-dependent genes in sperm, it is not surprising that in both experiments (mTORC1 and mTORC2 suppression) the lists of genes associated with significant KO- dependent DMRs were enriched with developmental categories (**Fig. 5C,D**), indicative that age- dependent changes in mTOR pathway and in the BTB permeability may be causative of the transfer of altered developmental epigenetic information to the next generation. Specifically, enriched categories common for both experiments included brain development, heart development, chordate embryonic development, pattern specification process, cell morphogenesis, muscle structure development, signaling by receptor tyrosine kinases, skeletal system development, sensory organ development, behavior, and protein phosphorylation. Thus, there is a high concordance of developmental categories enriched with genes associated with sperm methylation changes induced by natural aging (**Fig. 5A,B**) and induced by changes in mTOR complexes activity balance (**Fig. 5C,D**) (see also Dryad dataset (Amir et al., 2023)).

## Decreased mTORC2 activity is concordant with aged reproductive phenotype

We next analyzed if Sertoli-specific manipulation of the mTOR pathway may result in reproductive phenotypes concordant with age-related changes. It was demonstrated previously that testes weight and sperm counts decrease with age, while the percent of morphologically abnormal spermatozoa increases with age in humans and laboratory rodents (Bujan et al., 1988; Gunes et al., 2016). Additionally, mitochondria DNA copy number (mtDNAcn) increases with age in human sperm (Zhang et al., 2016) and it was shown recently that sperm mtDNAcn is a novel biomarker of male fecundity (Rosati et al., 2020; H. Wu et al., 2019). Thus, to analyze if manipulated mTOR pathway affects known age-dependent reproductive parameters, we analyzed testes size, sperm counts, sperm morphology and mtDNAcn in mice of all genotypes across four timepoints covering ages that are equivalent to critical reproductive stages in humans: 8 week old – young pubertal; 12 and 22 weeks old – mature adult; and 56 weeks old – advanced paternal age (Bell, 2018; Flurkey et al., 2007). Concordant with our hypothesis, mice with inactivated mTORC2 (permanently “leaky” BTB) demonstrated significant decreases in testis weight and sperm counts concomitant with increase in morphologically abnormal sperm and mtDNAcn (**Table 1**, **Fig. 6A-D**). Inactivation of mTORC1 resulted in testes weight decrease and slight increase in mtDNAcn but did not affect sperm parameters significantly. These results suggest that although



permanent changes of the mTOR complexes balance in either direction may have negative consequences for some reproductive parameters, mice with inactivated mTORC2 in Sertoli cells have a reproductive phenotype concordant with accelerated aging.

## Discussion

An accumulating body of literature indicates that the sperm epigenome responds to a broad range of factors, including paternal diet (Watkins et al., 2018 [\[1\]](#)), physical activity (Benito et al., 2018 [\[2\]](#)), stress (L. Wu et al., 2016 [\[3\]](#)), metabolic status (Donkin et al., 2016 [\[4\]](#); Wei et al., 2014 [\[5\]](#)), and environmental exposures (H. Wu et al., 2017 [\[6\]](#)), among others. Age is the most powerful known factor affecting DNA methylation in mammalian sperm (reviewed at (Ashapkin et al., 2022 [\[7\]](#))), and age-dependent changes in epigenetic mechanisms in sperm are involved in the regulation of major developmental pathways (Ashapkin et al., 2022 [\[7\]](#)). Importantly, advanced paternal age at conception programs offspring life span for shorter duration via non-Mendelian transfer of inheritable information by spermatozoa (Xie et al., 2018 [\[8\]](#)). Additionally, the rates of sperm epigenetic aging (DNA methylation and small noncoding RNA) may be modified by exposures to environmental xenobiotics (Pilsner et al., 2021 [\[9\]](#); Suvorov et al., 2020 [\[10\]](#)).

Despite this evidence, the mechanism(s) linking age or other paternal factors with epigenetic changes in sperm were unknown. The results of our experiments suggest that shifts of the balance of mTOR complexes activities in favor of mTORC1 or mTORC2 in Sertoli cells produce changes in sperm DNA methylation matching aging and rejuvenation respectively. Additionally, decreased activity of mTORC2 produced changes in reproductive parameters concordant with aging. Thus, mTOR pathway in Sertoli cells may be used as novel target of therapeutic interventions to rejuvenate the sperm epigenome in advanced-age fathers.

The complete cascade of molecular events that links mTOR complexes activity with rates of sperm epigenome aging remains unclear. The same mechanism, mTOR complexes balance in Sertoli cells, is responsible for BTB permeability regulation (Li & Cheng, 2016 [\[11\]](#); Mok, Mruk, & Cheng, 2013 [\[12\]](#); Mok, Mruk, Lee, et al., 2013). Given that BTB is responsible for the maintenance of a unique biochemical environment for germ cells undergoing DNA methylation changes (Gaysinskaya et al., 2018 [\[13\]](#); Loukinov et al., 2002 [\[14\]](#); Oakes et al., 2007 [\[15\]](#); Omri-Charai et al., 2023 [\[16\]](#)) and its permeability increases with age (Levy et al., 1999 [\[17\]](#); Paul & Robaire, 2013 [\[18\]](#)), these facts taken together provide a strong basis for a hypothesis that epigenetic aging of sperm results from the loss of BTB integrity. Additional, research is needed to test this hypothesis.

## Conflict of interest

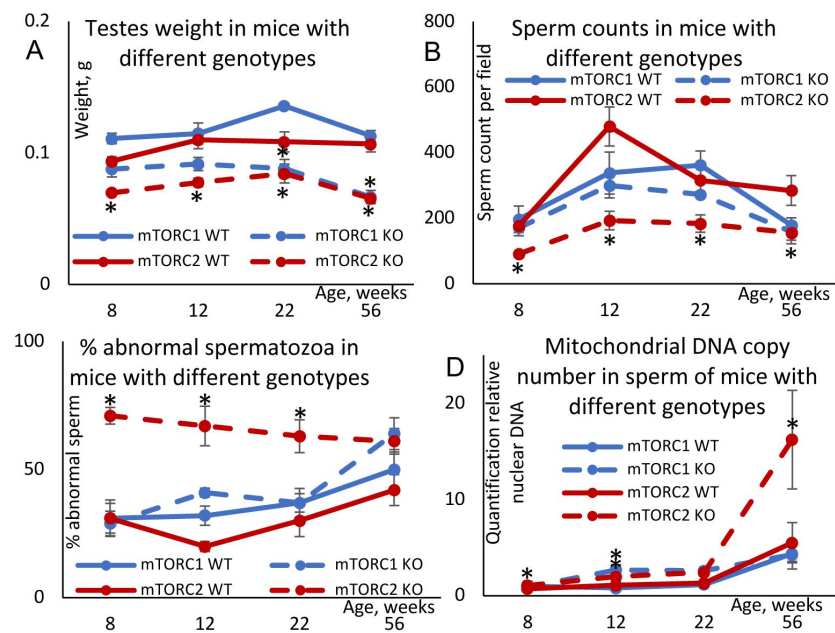
AS is a CEO of ReGENE LLC. All other authors declare that they have no conflict of interest.

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## Author contributions

Conceptualization: AS



**Fig. 6.**

**Changes in age-dependent reproductive parameters in mice with inactivated mTORC1 or mTORC2 in Sertoli cells.**

**(A)** Testes weight. **(B)** Sperm counts in sperm smears. **(C)** Percent abnormal spermatozoa. **(D)** Mitochondrial DNA copy number. All data are Mean  $\pm$  SE,  $n = 5-6$ /age/genotype, \* $q < 0.05$  when compared with corresponding WT.

| Experiment         | Genotype  | Age, weeks | Testes weight, mg          |                  | Sperm count/field            |                  | % abnormal spermatozoa     |              | Mitochondrial DNA copy number    |              |
|--------------------|-----------|------------|----------------------------|------------------|------------------------------|------------------|----------------------------|--------------|----------------------------------|--------------|
|                    |           |            | Mean $\pm$ SE              | q                | Mean $\pm$ SE                | q                | Mean $\pm$ SE              | q            | Mean $\pm$ SE                    | q            |
| mTORC1 suppression | WT        | 8          | 111 $\pm$ 4                | --               | 197 $\pm$ 40                 | --               | 31 $\pm$ 6                 | --           | 0.97 $\pm$ 0.53                  | --           |
|                    |           | 12         | 115 $\pm$ 8                | --               | 338 $\pm$ 64                 | --               | 32 $\pm$ 4                 | --           | 0.82 $\pm$ 0.10                  | --           |
|                    |           | 22         | 136 $\pm$ 2                | --               | 362 $\pm$ 43                 | --               | 37 $\pm$ 6                 | --           | 1.17 $\pm$ 0.26                  | --           |
|                    |           | 56         | 113 $\pm$ 4                | --               | 179 $\pm$ 22                 | --               | 50 $\pm$ 7                 | --           | 4.38 $\pm$ 0.82                  | --           |
|                    | Raptor KO | 8          | 88 $\pm$ 6                 | 0.062            | 171 $\pm$ 24                 | 0.725            | 29 $\pm$ 5                 | 0.867        | 0.93 $\pm$ 0.19                  | 0.307        |
|                    |           | 12         | 92 $\pm$ 5                 | 0.064            | 300 $\pm$ 38                 | 0.743            | 41 $\pm$ 2                 | 0.126        | <b>2.67<math>\pm</math>0.56</b>  | <b>0.024</b> |
|                    |           | 22         | <b>88<math>\pm</math>7</b> | <b>0.006</b>     | 272 $\pm$ 10                 | 0.169            | 37 $\pm$ 4                 | 0.950        | 2.60 $\pm$ 0.39                  | 0.056        |
|                    |           | 56         | <b>67<math>\pm</math>5</b> | <b>0.003</b>     | 159 $\pm$ 26                 | 0.728            | 64 $\pm$ 6                 | 0.568        | 4.28 $\pm$ 1.50                  | 0.665        |
| mTORC2 suppression | WT        | 8          | 93 $\pm$ 3                 | --               | 176 $\pm$ 6                  | --               | 31 $\pm$ 7                 | --           | 0.72 $\pm$ 0.03                  | --           |
|                    |           | 12         | 110 $\pm$ 7                | --               | 480 $\pm$ 60                 | --               | 20 $\pm$ 2                 | --           | 1.10 $\pm$ 0.19                  | --           |
|                    |           | 22         | 108 $\pm$ 7                | --               | 316 $\pm$ 41                 | --               | 30 $\pm$ 6                 | --           | 1.30 $\pm$ 0.10                  | --           |
|                    |           | 56         | 107 $\pm$ 7                | --               | 285 $\pm$ 45                 | --               | 42 $\pm$ 6                 | --           | 5.52 $\pm$ 2.08                  | --           |
|                    | Rictor KO | 8          | <b>70<math>\pm</math>1</b> | <b>0.050</b>     | <b>91<math>\pm</math>3</b>   | <b>0.001</b>     | <b>71<math>\pm</math>3</b> | <b>0.005</b> | <b>1.07<math>\pm</math>0.12</b>  | <b>0.017</b> |
|                    |           | 12         | <b>78<math>\pm</math>3</b> | <b>0.009</b>     | <b>193<math>\pm</math>28</b> | <b>0.009</b>     | <b>67<math>\pm</math>8</b> | <b>0.001</b> | <b>1.99<math>\pm</math>0.43</b>  | <b>0.016</b> |
|                    |           | 22         | <b>84<math>\pm</math>7</b> | <b>0.048</b>     | <b>184<math>\pm</math>26</b> | <b>0.049</b>     | <b>63<math>\pm</math>6</b> | <b>0.016</b> | 2.45 $\pm$ 0.34                  | 0.143        |
|                    |           | 56         | <b>66<math>\pm</math>2</b> | <b>&lt;0.001</b> | <b>156<math>\pm</math>34</b> | <b>&lt;0.001</b> | 61 $\pm$ 5                 | 0.711        | <b>16.23<math>\pm</math>5.12</b> | <b>0.049</b> |

**Table 1.**

Changes in reproductive parameters induced by Sertoli cell specific KO of Raptor (mTORC1 suppression) and Rictor (mTORC2 suppression). All parameters significantly different ( $q \leq 0.05$ ) in KO mice as compared with same age WT counterparts are shown in bold.

Methodology: SA, OA, EM, JM, JRP, OO, OS, AS

Software: SA, OA, EM, OO Validation: SA, OA

Formal analysis: SA, OA, EM

Investigation: SA, OA, EM, JM, JRP, OO, OS, AS

Resources: AS, JRP

Data Curation: SA, OA

Visualization: SA, OA, EM, JM, OO

Funding acquisition: AS

Project administration: AS Supervision: JRP, AS

Writing – original draft: AS

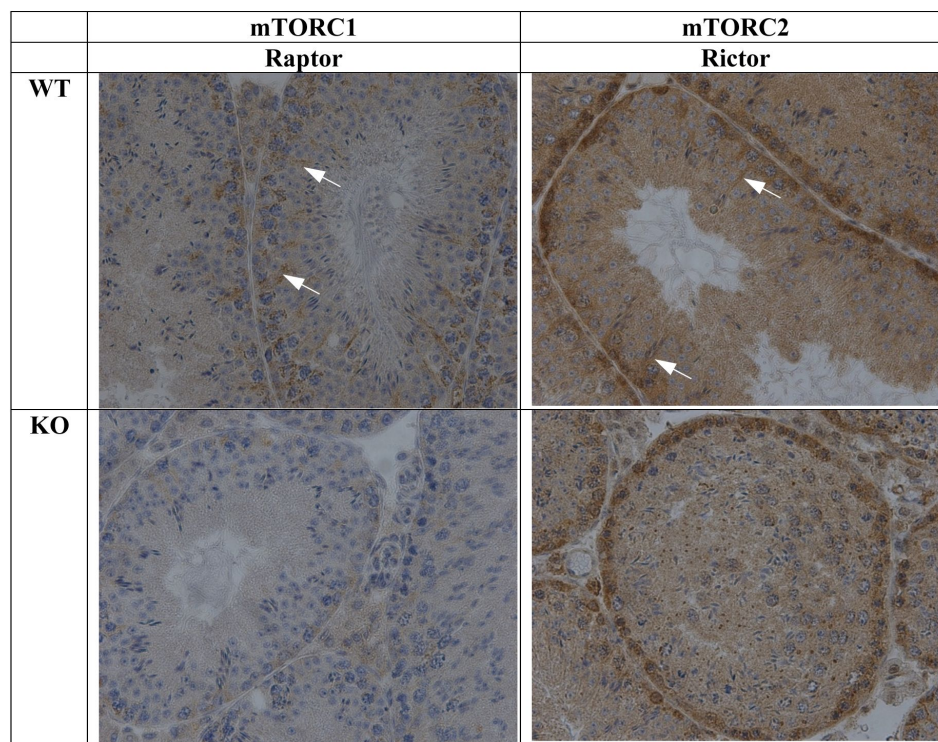
Writing – review & editing: SA, OA, EM, JM, JRP, OO, OS, AS

## Data availability

Raw and processed DNA methylation data as well as complete results of Metascape enrichment analysis are available via Dryad ([Amir et al., 2023](#)). All other data are available in the main text or the Extended Data.

## Ethics statement

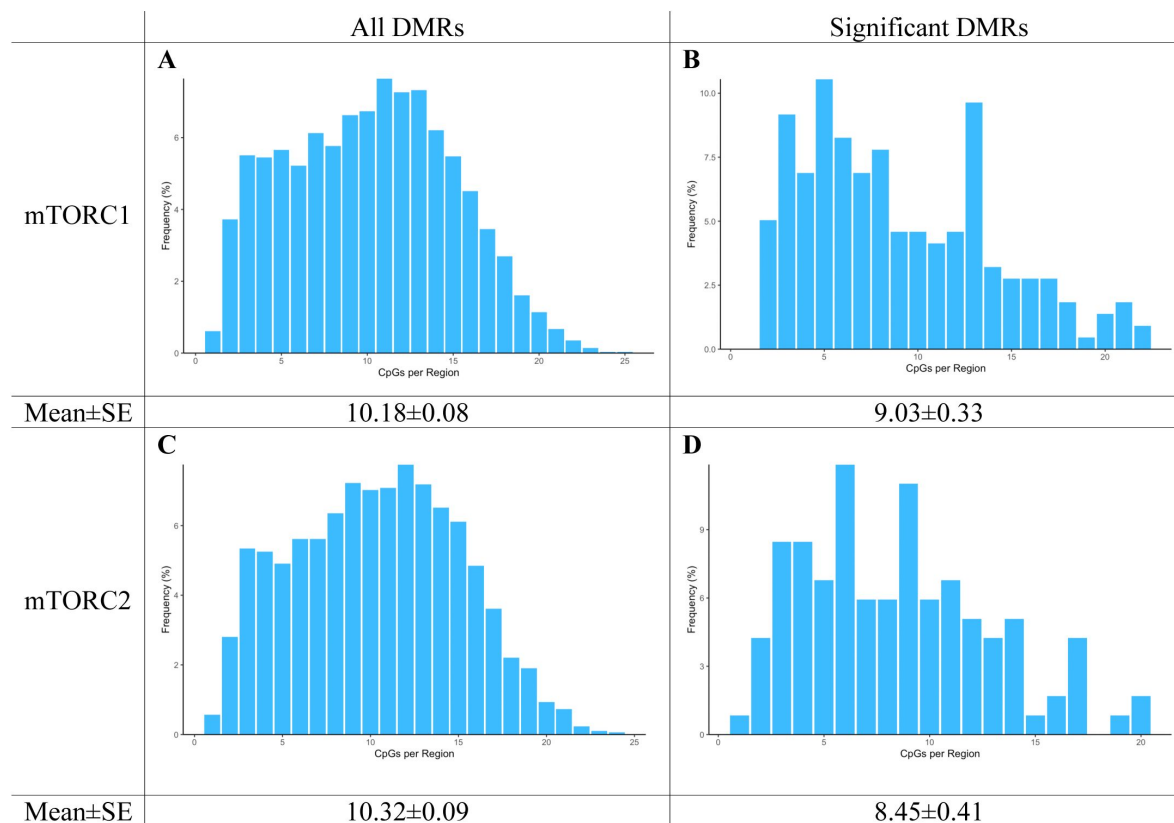
All procedures followed the guidelines of the National Institutes of Health Guide for the Care and Use of Laboratory Animals and the approval for this study was received from the Institutional Animal Care and Use Committee at University of Massachusetts, Amherst.



**Fig. S1.**

**Immunohistochemistry analysis of the efficiency of Rictor and Raptor KO in Sertoli cells.**

Representative images illustrate lack of staining for Rictor and Raptor proteins in Sertoli cells in corresponding KO models. Arrows show staining by Rictor or Raptor antibodies respectively in Sertoli cells.

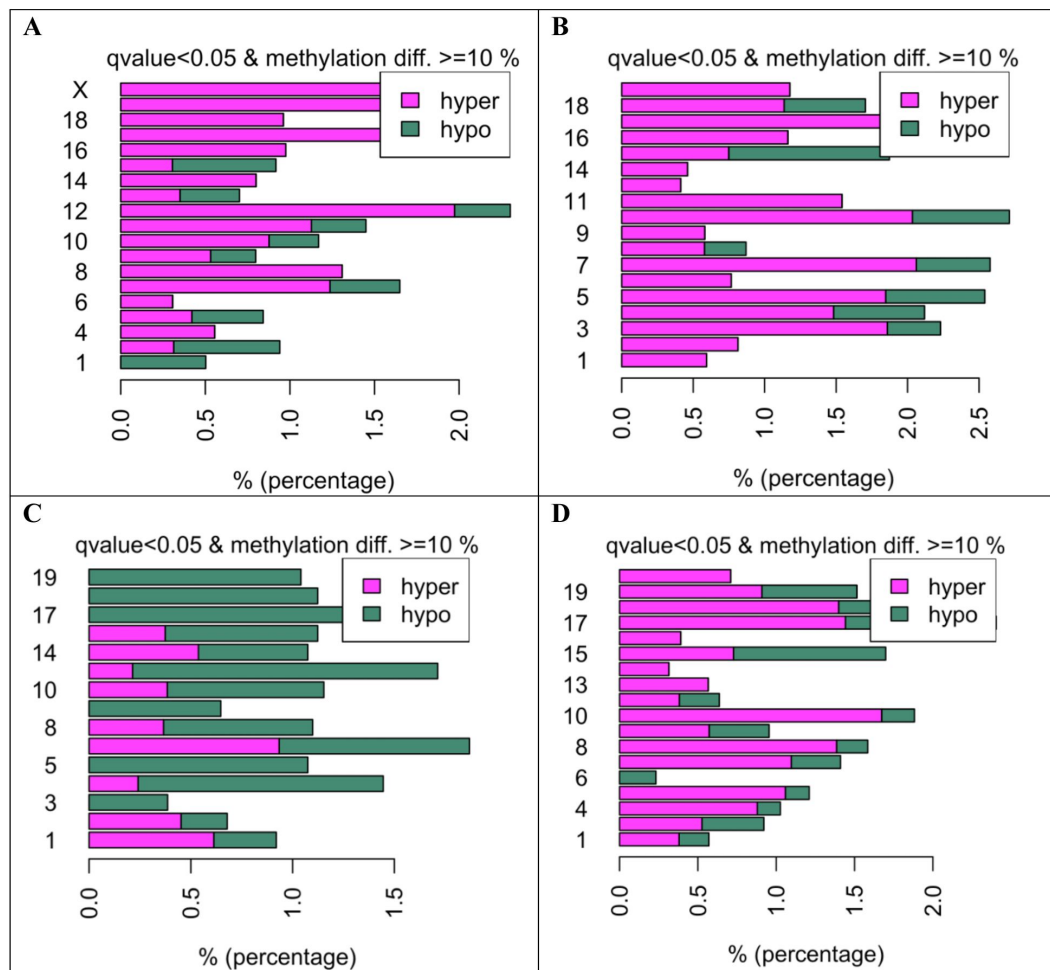


**Fig. S2.**

**Distribution of CpG per 100 bp regions.**

(A) All overlapping regions identified in age and genotype groups in mTORC1 inactivation experiment. (B) Significant ( $q < 0.05$ ) overlapping regions identified in age and genotype groups in mTORC1 inactivation experiment. (C) All overlapping regions identified in age and genotype groups in mTORC2 inactivation experiment. (D) Significant ( $q < 0.05$ ) overlapping regions identified in age and genotype groups in mTORC2 inactivation experiment.

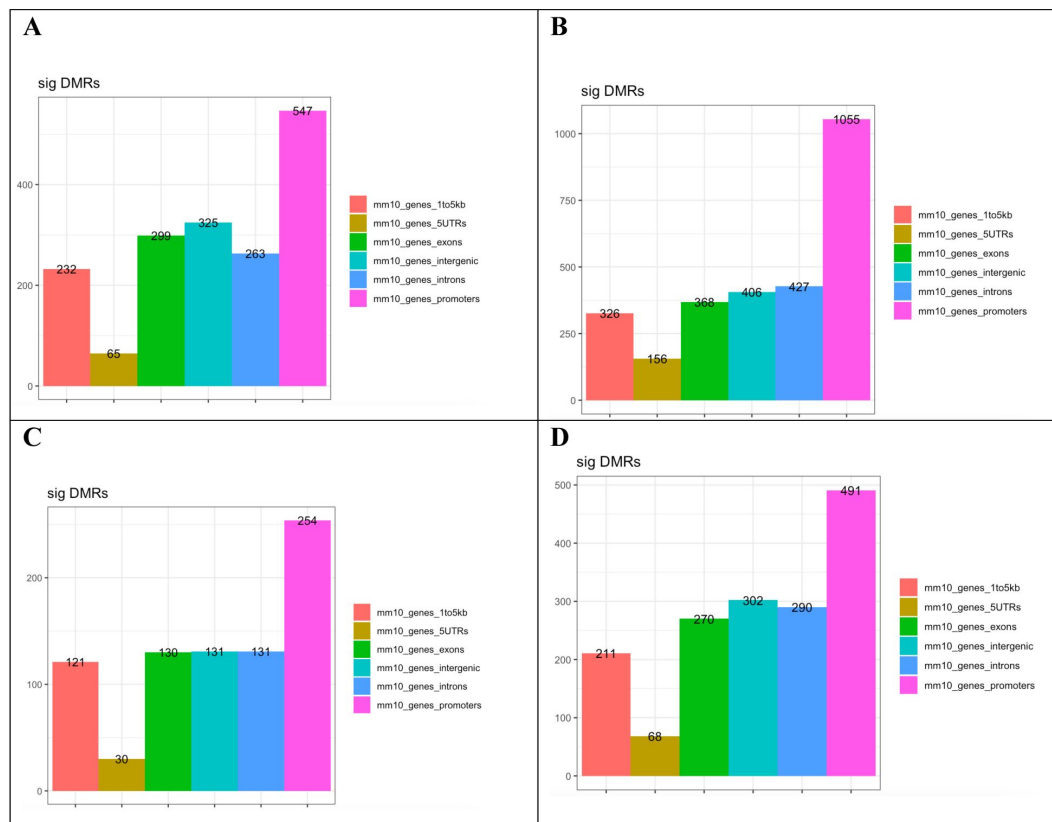




**Fig. S3.**

### Methylation changes per chromosome in mice of different genotype and age.

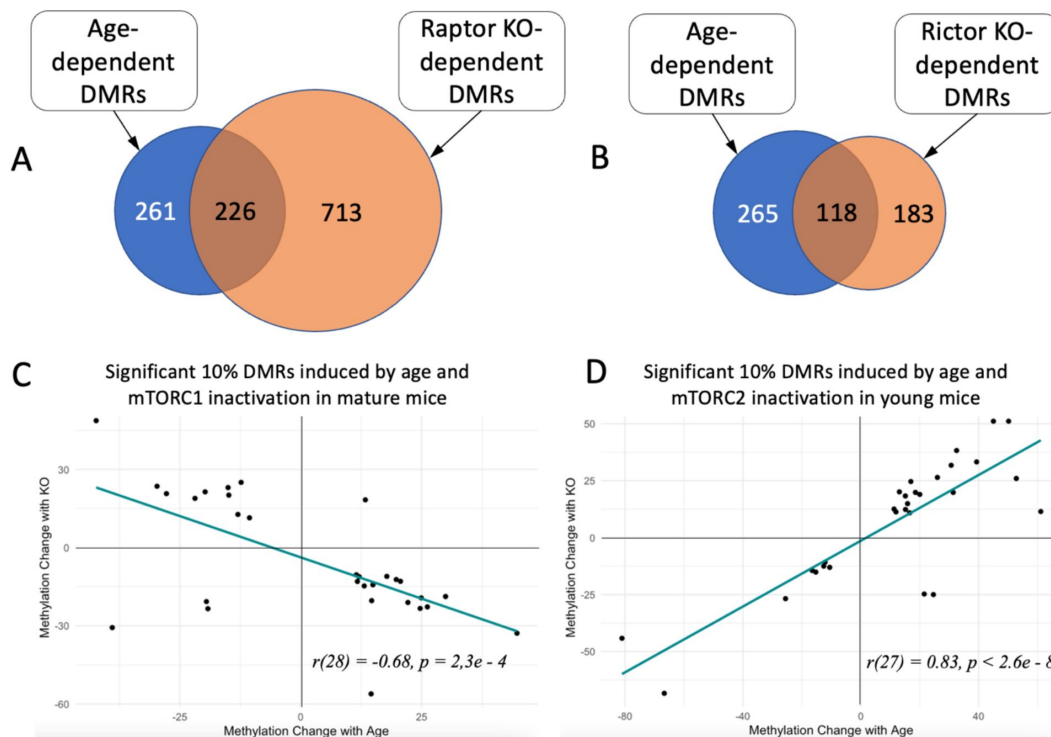
**(A and B).** Percent DMRs altered by age in WT animals in mTORC1 **(A)** and mTORC2 **(B)** inactivation experiments. **(C)** Percent DMRs in mature mice altered by mTORC1 inactivation. **(D)** Percent DMRs in young mice altered by mTORC2 inactivation.



**Fig. S4.**

**Genomic elements enriched with DMRs associated with age and inactivation of mTOR complexes.**

Changes induced by age in WT animals in mTORC1 (**A**) and mTORC2 (**B**) inactivation experiments. (**C**) Changes induced in mature mice by mTORC1 inactivation. (**D**) Changes induced in young mice by mTORC2 inactivation.



**Fig. S5.**

### Changes in methylation of significant DMRs induced by age and genetic manipulation of the mTOR pathway in Sertoli cells.

**(A)** Overlap between significant DMRs induced by age and by mTORC1 inactivation in mature mice among all common methylation regions identified in both comparisons. **(B)** Overlap between significant DMRs induced by age and by mTORC2 inactivation in young mice among all common methylation regions identified in both comparisons. **(C)** Comparison of age-dependent changes and changes induced by mTORC1 inactivation in mature mice. **(D)** Comparison of age-dependent changes and changes induced by mTORC2 inactivation in young mice.

## References

- Amir S., Arowolo O., Mironova Ekaterina, McGaunn J., Oluwayiose O. A., Sergeyev Oleg, Pilsner J. R., Suvorov A (2023) **Mouse sperm DNA-methylation changes induced by age and mechanistic target of rapamycin (mTOR) manipulation in Sertoli cells [dataset]** *Dryad* <https://doi.org/10.5061/dryad.ncjsxkt0m>
- Ashapkin V., Suvorov A., Pilsner J., Krawetz S., Sergeyev O (2022) **Age-associated epigenetic changes in mammalian sperm: Implications for offspring health and development** *Human Reproduction Update* **29**:24–44 <https://doi.org/10.1093/humupd/dmac033>
- Atsem S. *et al.* (2016) **Paternal age effects on sperm FOXX1 and KCNA7 methylation and transmission into the next generation.** *Human Molecular Genetics Journal Article*
- Bell M. R (2018) **Comparing Postnatal Development of Gonadal Hormones and Associated Social Behaviors in Rats, Mice, and Humans** *Endocrinology* **159**:2596–2613 <https://doi.org/10.1210/en.2018-00220>
- Benito E. *et al.* (2018) **RNA- Dependent Intergenerational Inheritance of Enhanced Synaptic Plasticity after Environmental Enrichment** *Cell Reports* **23**:546–554 <https://doi.org/10.1016/j.celrep.2018.03.059>
- Bray I., Gunnell D., Davey Smith G (2006) **Advanced paternal age: How old is too old?** *Journal of Epidemiology and Community Health* **60**:851–853
- Bujan L., Mieuisset R., Mondinat C., Mansat A., Pontonnier F (1988) **Sperm morphology in fertile men and its age related variation** *Andrologia* **20**:121–128 <https://doi.org/10.1111/j.1439-0272.1988.tb00672.x>
- Contreras Z. A., Hansen J., Ritz B., Olsen J., Yu F., Heck J. E (2017) **Parental age and childhood cancer risk: A Danish population-based registry study** *Cancer Epidemiology* **49**:202–215 <https://doi.org/10.1016/j.canep.2017.06.010>
- Denomme M. M., Haywood M. E., Parks J. C., Schoolcraft W. B., Katz-Jaffe M. G (2020) **The inherited methylome landscape is directly altered with paternal aging and associated with offspring neurodevelopmental disorders** *Aging Cell* **19** <https://doi.org/10.1111/accel.13178>
- Donkin I. *et al.* (2016) **Obesity and Bariatric Surgery Drive Epigenetic Variation of Spermatozoa in Humans** *Cell Metabolism* **23**:369–378 <https://doi.org/10.1016/j.cmet.2015.11.004>
- Du P., Zhang X., Huang C.-C., Jafari N., Kibbe W. A., Hou L., Lin S. M (2010) **Comparison of Beta-value and M-value methods for quantifying methylation levels by microarray analysis** *BMC Bioinformatics* **11** <https://doi.org/10.1186/1471-2105-11-587>
- Ernst J., Bar-Joseph Z (2006) **STEM: A tool for the analysis of short time series gene expression data** *BMC Bioinformatics* **7** <https://doi.org/10.1186/1471-2105-7-191>

- Fagerberg L. *et al.* (2014) **Analysis of the human tissue-specific expression by genome-wide integration of transcriptomics and antibody-based proteomics** *Molecular & Cellular Proteomics : MCP* **13**:397–406 <https://doi.org/10.1074/mcp.M113.035600>
- Flurkey K., Curren J., Harrison D (2007) **Mouse models in aging research** *Faculty Research* :637–672
- Gaysinskaya V., Miller B. F., De Luca C., van der Heijden G. W., Hansen K. D., Bortvin A. (2018) **Transient reduction of DNA methylation at the onset of meiosis in male mice** *Epigenetics & Chromatin* **11** <https://doi.org/10.1186/s13072-018-0186-0>
- Gratten J., Wray N. R., Peyrot W. J., McGrath J. J., Visscher P. M., Goddard M. E (2016) **Risk of psychiatric illness from advanced paternal age is not predominantly from de novo mutations** *Nature Genetics* **48** <https://doi.org/10.1038/ng.3577>
- Gunes S., Hekim G. N. T., Arslan M. A., Asci R (2016) **Effects of aging on the male reproductive system** *Journal of Assisted Reproduction and Genetics* **33**:441–454 <https://doi.org/10.1007/s10815-016-0663-y>
- Guo Y. *et al.* (2021) **Altered sperm tsRNAs in aged male contribute to anxiety-like behavior in offspring** *Aging Cell* **20** <https://doi.org/10.1111/accel.13466>
- Hassan M. A., Killick S. R (2003) **Effect of male age on fertility: Evidence for the decline in male fertility with increasing age** *Fertility and Sterility* **79**:1520–1527
- Horta F., Vollenhoven B., Healey M., Busija L., Catt S., Temple-Smith P (2019) **Male ageing is negatively associated with the chance of live birth in IVF/ICSI cycles for idiopathic infertility** *Human Reproduction* **34**:2523–2532 <https://doi.org/10.1093/humrep/dez223>
- Huber M. H., Bronson F. H., Desjardins C (1980) **Sexual activity of aged male mice: Correlation with level of arousal, physical endurance, pathological status, and ejaculatory capacity** *Biology of Reproduction* **23**:305–316 <https://doi.org/10.1095/biolreprod23.2.305>
- Jenkins T. G., Aston K. I., Pflueger C., Cairns B. R., Carrell D. T (2014) **Age-associated sperm DNA methylation alterations: Possible implications in offspring disease susceptibility** *PLoS Genetics* **10** <https://doi.org/10.1371/journal.pgen.1004458>
- Leek J. T., Johnson W. E., Parker H. S., Jaffe A. E., Storey J. D (2012) **The sva package for removing batch effects and other unwanted variation in high-throughput experiments** *Bioinformatics* **28**:882–883 <https://doi.org/10.1093/bioinformatics/bts034>
- Levy S., Serre V., Hermo L., Robaire B (1999) **The effects of aging on the seminiferous epithelium and the blood-testis barrier of the Brown Norway rat** *Journal of Andrology* **20**:356–365
- Li N., Cheng C. Y (2016) **Mammalian target of rapamycin complex (mTOR) pathway modulates blood-testis barrier (BTB) function through F-actin organization and gap junction** *Histology and Histopathology* **31**:961–968 <https://doi.org/10.14670/HH-11-753>

- Lisner A., Kimmins S (2023) **Emerging evidence that the mammalian sperm epigenome serves as a template for embryo development** *Nature Communications* **14** <https://doi.org/10.1038/s41467-023-37820-2>
- Loukinov D. I. *et al.* (2002) **BORIS, a novel male germ-line-specific protein associated with epigenetic reprogramming events, shares the same 11-zinc-finger domain with CTCF, the insulator protein involved in reading imprinting marks in the soma** *Proceedings of the National Academy of Sciences of the United States of America* **99**:6806–6811 <https://doi.org/10.1073/pnas.092123699>
- Ma J. *et al.* (2020) **Mitochondria-related miR- 574 reduces sperm ATP by targeting ND5 in aging males** *Aging* **12**:8321–8338 <https://doi.org/10.18632/aging.103141>
- Meng J., Holdcraft R. W., Shima J. E., Griswold M. D., Braun R. E (2005) **Androgens regulate the permeability of the blood-testis barrier** *Proceedings of the National Academy of Sciences of the United States of America* **102**:16696–16700
- Mills M., Rindfuss R. R., McDonald P., te Velde E., on behalf of the ESHRE Reproduction and Society Task Force (2011) **Why do people postpone parenthood? Reasons and social policy incentives** *Human Reproduction Update* **17**:848–860 <https://doi.org/10.1093/humupd/dmr026>
- Mochida K., Hasegawa A., Ogonuki N., Inoue K., Ogura A (2019) **Early production of offspring by in vitro fertilization using first-wave spermatozoa from prepubertal male mice** *Journal of Reproduction and Development* **65**:467–473 <https://doi.org/10.1262/jrd.2019-042>
- Mok K. W., Mruk D. D., Cheng C. Y (2013) **Regulation of blood-testis barrier (BTB) dynamics during spermatogenesis via the “Yin” and “Yang” effects of mammalian target of rapamycin complex 1 (mTORC1) and mTORC2** *International Review of Cell and Molecular Biology* **301**:291–358 <https://doi.org/10.1016/B978-0-12-407704-1.00006-3>
- Mok K. W., Mruk D. D., Lee W. M., Cheng C. Y (2013) **Rictor/mTORC2 regulates blood- testis barrier dynamics via its effects on gap junction communications and actin filament network** *FASEB Journal : Official Publication of the Federation of American Societies for Experimental Biology* **27**:1137–1152 <https://doi.org/10.1096/fj.12-212977>
- Montgomery S. M., Lambe M., Olsson T., Ekblom A (2004) **Parental age, family size, and risk of multiple sclerosis** *Epidemiology* **15**:717–723 <https://doi.org/10.1097/01.ede.0000142138.46167.69>
- Niu L., Xu Z., Taylor J. A (2016) **RCP: A novel probe design bias correction method for Illumina Methylation BeadChip** *Bioinformatics* **32**:2659–2663 <https://doi.org/10.1093/bioinformatics/btw285>
- Oakes C. C., La Salle S., Smiraglia D. J., Robaire B., Trasler J. M. (2007) **Developmental acquisition of genome-wide DNA methylation occurs prior to meiosis in male germ cells** *Developmental Biology* **307**:368–379
- Oluwayiose O. A., Marcho C., Wu H., Houle E., Krawetz S. A., Suvorov A., Mager J., Richard Pilsner J (2021) **Paternal preconception phthalate exposure alters sperm methylome and embryonic programming** *Environment International* **155** <https://doi.org/10.1016/j.envint.2021.106693>



- Omri-Charai R. E., Gilbert I., Prunier J., Desmarais R., Ghinet M. G., Robert C., Boissonneault G., Delbes G (2023) **DNA methylation dynamic in male rat germ cells during gametogenesis** *Development (Cambridge, England), dev* **201606** <https://doi.org/10.1242/dev.201606>
- Papadopoli D., Boulay K., Kazak L., Pollak M., Mallette F., Topisirovic I., Hulea L (2019) **mTOR as a central regulator of lifespan and aging** *F1000Research* **8** <https://doi.org/10.12688/f1000research.17196.1>
- Paul C., Robaire B (2013) **Impaired function of the blood-testis barrier during aging is preceded by a decline in cell adhesion proteins and GTPases** *PLoS One* **8** <https://doi.org/10.1371/journal.pone.0084354>
- Pilsner J. R., Shershebnayev A., Wu H., Marcho C., Dribnokhodova O., Shtratnikova V., Sergeyev O., Suvorov A (2021) **Aging-induced changes in sperm DNA methylation are modified by low dose of perinatal flame retardants.** *Epigenomics Journal Article* <https://doi.org/10.2217/epi-2020-0404>
- Puleo C. M., Schmeidler J., Reichenberg A., Kolevzon A., Soorya L. V., Buxbaum J. D., Silverman J. M (2012) **Advancing paternal age and simplex autism** *Autism* **16**:367–380 <https://doi.org/10.1177/1362361311427154>
- Reichenberg A. *et al.* (2006) **Advancing paternal age and autism** *Archives of General Psychiatry* **63**:1026–1032
- Rosati A. J., Whitcomb B. W., Brandon N., Buck Louis G. M., Mumford S. L., Schisterman E. F., Pilsner J. R (2020) **Sperm mitochondrial DNA biomarkers and couple fecundity** *Human Reproduction* **35**:2619–2625 <https://doi.org/10.1093/humrep/deaa191>
- Saha S., Barnett A. G., Foldi C., Burne T. H., Eyles D. W., Buka S. L., McGrath J. J (2009) **Advanced paternal age is associated with impaired neurocognitive outcomes during infancy and childhood** *PLoS Medicine* **6** <https://doi.org/10.1371/journal.pmed.1000040>
- Suvorov A (2021) **Population Numbers and Reproductive Health** *Endocrinology* **162** <https://doi.org/10.1210/endocr/bqab154>
- Suvorov A., Pilsner J. R., Naumov V., Shtratnikova V., Zheludkevich A., Gerasimov E., Logacheva M., Sergeyev O (2020) **Aging Induces Profound Changes in sncRNA in Rat Sperm and These Changes Are Modified by Perinatal Exposure to Environmental Flame Retardant** *International Journal of Molecular Sciences* **21** <https://doi.org/10.3390/ijms21218252>
- Waldenström U (2016) **Postponing parenthood to advanced age** *Upsala Journal of Medical Sciences* **121**:235–243 <https://doi.org/10.1080/03009734.2016.1201553>
- Watkins A. J., Dias I., Tsuro H., Allen D., Emes R. D., Moreton J., Wilson R., Ingram R. J. M., Sinclair K. D (2018) **Paternal diet programs offspring health through sperm- and seminal plasma-specific pathways in mice** *Proceedings of the National Academy of Sciences of the United States of America* **115**:10064–10069 <https://doi.org/10.1073/pnas.1806333115>
- Wei Y., Yang C. R., Wei Y. P., Zhao Z. A., Hou Y., Schatten H., Sun Q. Y (2014) **Paternally induced transgenerational inheritance of susceptibility to diabetes in mammals** *Proceedings of the National Academy of Sciences of the United States of America* **111**:1873–1878 <https://doi.org/10.1073/pnas.1321195111>

- Weichhart T (2018) **mTOR as regulator of lifespan, aging and cellular senescence** *Gerontology* **64**:127–134 <https://doi.org/10.1159/000484629>
- Wu C., Blondin P., Vigneault C., Labrecque R., Sirard M. A (2020) **Sperm miRNAs- potential mediators of bull age and early embryo development** *BMC Genomics* **21**:798–20 <https://doi.org/10.1186/s12864-020-07206-5>
- Wu H., Estill M. S., Shershebnov A., Suvorov A., Krawetz S. A., Whitcomb B. W., Dinnie H., Rahil T., Sites C. K., Pilsner J. R (2017) **Preconception urinary phthalate concentrations and sperm DNA methylation profiles among men undergoing IVF treatment: A cross-sectional study** *Human Reproduction* **32**:2159–2169 <https://doi.org/10.1093/humrep/dex283>
- Wu H., Hauser R., Krawetz S. A., Pilsner J. R. (2015) **Environmental Susceptibility of the Sperm Epigenome During Windows of Male Germ Cell Development** *Current Environmental Health Reports, Journal Article* <https://doi.org/10.1007/s40572-015-0067-7>
- Wu H., Whitcomb B. W., Huffman A., Brandon N., Labrie S., Tougias E., Lynch K., Rahil T., Sites C. K., Pilsner J. R (2019) **Associations of sperm mitochondrial DNA copy number and deletion rate with fertilization and embryo development in a clinical setting** *Human Reproduction* **34**:163–170 <https://doi.org/10.1093/humrep/dey330>
- Wu L. *et al.* (2016) **Paternal Psychological Stress Reprograms Hepatic Gluconeogenesis in Offspring** *Cell Metabolism* **23**:735–743 <https://doi.org/10.1016/j.cmet.2016.01.014>
- Xie K. *et al.* (2018) **Epigenetic alterations in longevity regulators, reduced life span, and exacerbated aging-related pathology in old father offspring mice** *Proceedings of the National Academy of Sciences of the United States of America* **115**:E2348–E2357 <https://doi.org/10.1073/pnas.1707337115>
- Yoshizaki K. *et al.* (2021) **Paternal age affects offspring via an epigenetic mechanism involving REST/NRSF** *EMBO Reports* **22** <https://doi.org/10.15252/embr.202051524>
- Yue F. *et al.* (2014) **A comparative encyclopedia of DNA elements in the mouse genome** *Nature* <https://doi.org/10.1038/nature13992>
- Zhang G. *et al.* (2016) **Mitochondrial Biomarkers Reflect Semen Quality: Results from the MARCHS Study in Chongqing, China** *PloS One* **11** <https://doi.org/10.1371/journal.pone.0168823>
- Zhou Y., Zhou B., Pache L., Chang M., Khodabakhshi A. H., Tanaseichuk O., Benner C., Chanda S. K (2019) **Metascope provides a biologist-oriented resource for the analysis of systems-level datasets** *Nature Communications* **10**:1523–19 <https://doi.org/10.1038/s41467-019-09234-6>

## Editors

Reviewing Editor

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ETH Zurich, Zurich, Switzerland

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**Reviewer #1 (Public Review):**

In the manuscript "Mechanistic target of rapamycin (mTOR) pathway in Sertoli cells regulates age-dependent changes in sperm DNA methylation", the authors proposed to test if the balance of mTOR complexes in Sertoli cells may play a significant role in age-dependent changes in the sperm epigenome. The paper could be of interest and has a good scientific aim but there are too many drawbacks that hamper the initial enthusiasm. All sections need extensive revision. The paper is mostly descriptive without a mechanistic-orientated explanation for the observed results.

Comments on revised version:

I am not sure that the authors have made an attempt to clearly answer the reviewers comments that aimed to improve the quality of the manuscript. It stands as mostly descriptive and with limited interest as it is.

<https://doi.org/10.7554/eLife.90992.2.sa2>

**Reviewer #3 (Public Review):**

Summary and Strength:

The manuscript by Amir et al. describes that Sertoli-specific inactivation of the mTORC1 and mTORC2 complex by KO of either Raptor or Rictor, respectively, resulted in progressive changes in blood-testis-barrier (BTB) function, testis weight, and sperm parameters, including counts, morphology, mtDNA content and sperm DNA methylation.

The described studies are based on the hypothesis that a decline of BTB function with increasing chronological age of a male contributes to the DNA methylation changes that are known to occur in sperm DNA of old males when compared to sperm DNA from isogenic young males. In order to demonstrate the relevance of a functioning BTB for the maintenance of sperm methylation patterns, the authors generated mice with genetically disrupted mTORC2 complex or mTORC1 complex in Sertoli cells and determined sperm methylation patterns in comparison to isogenic wild-type males. In line with previously published scientific literature (e.g. Mok et al., 2013; Dong et al, 2015; and others), the manuscript corroborates that a Sertoli-cell specific deletion of mTORC2 caused a loss of BTB function and a progressive spermatogenic defect. The authors further show that sperm DNA is differentially methylated (DMRs) as a consequence of either a mTORC2 disruption (associated with a loss of BTB function) or following a mTORC1 disruption (BTB function either increased or not leaky) when compared to their isogenic age-matched wt controls. Those DMRs overlap partially with changes in sperm DNA methylation that were found when comparing sperm from 8-week males with sperm isolated from 22-week-old male mice.

The authors interpret the observed changes as representative of the sperm DNA methylation changes that occur during normal chronological aging of the male. For an aged control group, the authors use sperm DNA of 22-week-old wild-type males from the mTORC2 and mTORC2 KO breeding and compare the sperm methylation patterns found in sperm from those 22-week males to 8-week young males, that are intended to represent an old and a young cohort, respectively. DNA methylation analysis indicates that a disruption of mTORC2 (& decrease of BTB function) results in increased DNA methylation of sperm DNA, while a disruption of mTORC1 (and proposed increase of BTB tightness, not shown in the manuscript, though) resulted in increased hypomethylation.

Weaknesses:

While the hypothesis and experimental system are interesting and the data demonstrating the relevance of the mTORC2 complex for BTB function is convincing, several open questions limit the evidence that supports the hypothesis that the sperm DNA methylation changes seen in old males are caused by BTB failure following an imbalance of mTOR signaling complexes. The major critique points are the lack of a chronologically old group and the choice of 8 weeks & 22 weeks age of age:

- Data illustrating the degree of BTB decline and sperm DNA methylation changes from chronologically "old" male mice is missing. 22-week-old mice are not considered old but are of good and mature breeding age, equivalent to humans in their mid-late twenties. (In the manuscript, the 22-week-old wildtype mice show no evidence of BTB breakdown (Figure 3), so why are their sperm used to represent "aged" sperm?
- Adding a group of "old" wild-type mice of 12-14 months of age, which is closer to the end of effective reproduction in mice, more equivalent to 45-59 year-old humans) could be used to illustrate that (a) aging causes a marked decrease in BTB function at this time in mouse life, and that this BTB breakdown chronologically aligns with the age-associated DNA hypermethylation seen in old sperm. Age-matched "old" mTORC1 KO, with a (supposedly) tighter BTB barrier, could then be expected to have a sperm DNA methylation profile closer to that of younger wild-type animals. Such data are currently missing. While the progressive testicular decline observed in the mTORC1 KO (Fig.5) could make it difficult to obtain the appropriately aged mTORC1 KO tissues, it is completely feasible to obtain data from chronologically old wild-type males. (The progressive testicular decline further raises the question of what additional defects the KO causes, and how such additional defects would influence the sperm DNA methylation profile.) The addition of data from an old group to the currently included groups could strengthen the interpretation that the observations in the BTB-defective mTORC2 KO mice are modelling an age-related testicular decline, provided that the DMRs seen in the chronologically old group significantly overlap with the BTB-defective changes.
- In the current form, the described differences in sperm DNA methylation are based on comparisons between pubertal mice (8 weeks) and mature but not old adult males (22 weeks), while a chronologically "old" group is missing from the data sets and comparisons. Thus, it appears that the described sperm methylation changes reflect developmental changes associated with normal maturation and not necessarily declining sperm quality due to aging. (Sperm obtained from 8-week-old mice likely were generated, at least in part, during the 1st wave of spermatogenesis, which is known to differ from the continuously proceeding spermatogenesis during the remainder of the mature life. During the 1st wave of spermatogenesis, Sertoli cells are known to undergo gene expression changes which could contribute to varying degrees of BTB function, and thus have effects on the sperm DNA methylation profiles of such 1st wave sperm.)
- It is unclear why the aging-related DMRs between the 8 and 22-week-old wild-type mice vary so dramatically between the two wild-type groups derived from the mTORC1 and the mTORC2 breeding (Fig. S4). If the main difference was due to mTORC1 or mTORC2 activity, both wildtype groups should behave very similarly. Changes seen in a truly "old" mouse (e.g. 20 weeks to 56 weeks), changes in "young mTORC1" and in "old mTORC2" are missing. How do those numbers and profiles compare to the shown samples?

Comments on latest version:

The rebuttal letter and public response indicate the authors' reluctance to consider the limitations of their study, i.e. having chosen chronologically young animals to demonstrate a sperm aging effect and indicate that they are not willing to include adequate controls.

Since there is no evidence that mice at this young age have a deteriorating blood-testis-barrier (indeed, normal intact BTB is clearly visible in the figures included in this study from animals of the relevant age group), the whole central hypothesis that the study is built upon (i.e. that increasing age causes deteriorating BTB integrity which in turn causes age-related changes in sperm DNA methylation), appears irrelevant or invalid.

The authors' claim that age-related DNA methylation changes in sperm occur in linear fashion and that the changes are somewhat proportional with chronological age is in stark contrast of the claim that a decline of the BTB in old animals is causative for age-related sperm epigenetic changes, putting the relevance of the whole study in question.

<https://doi.org/10.7554/eLife.90992.2.sa1>

#### Author response:

The following is the authors' response to the current reviews.

##### **Reviewer #1 (Public Review):**

*In the manuscript "Mechanistic target of rapamycin (mTOR) pathway in Sertoli cells regulates age-dependent changes in sperm DNA methylation", the authors proposed to test if the balance of mTOR complexes in Sertoli cells may play a significant role in age-dependent changes in the sperm epigenome. The paper could be of interest and has a good scientific aim but there are too many drawbacks that hamper the initial enthusiasm. All sections need extensive revision. The paper is mostly descriptive without a mechanistic-orientated explanation for the observed results.*

##### *Comments on revised version:*

*I am not sure that the authors have made an attempt to clearly answer the reviewers comments that aimed to improve the quality of the manuscript. It stands as mostly descriptive and with limited interest as it is.*

We are thankful to the reviewer for agreeing to review our revised manuscript. Unfortunately, we completely disagree with the evaluation provided by the reviewer. Research on sperm DNA methylation experienced a significant rise of interest in the current century and by now more than 2000 papers have been published. Although it was demonstrated that the sperm DNA methylome may be affected by almost every factor analyzed, no study was published to identify molecular mechanisms that may link these factors with the sperm epigenome. Our study is the FIRST to identify such a mechanism (mTOR complexes balance in Sertoli cells). More so, we demonstrated experimentally that manipulations of this mechanism allow regulation of the rates of epigenetic aging of sperm in both directions (accelerate aging or rejuvenate). Thus, our study provides a mechanistic background for the development of therapeutic interventions that may target sperm epigenome.

We acknowledge that our study does not provide the full cascade of events linking the balance of mTOR complexes in Sertoli cells with the sperm DNA methylome. It suggests, however, the most plausible event next in a cascade (BTB permeability changes). Our group is working on this question now and we hope to provide the answer soon in a separate study. Even after that, we will be far from understanding the complete chain of molecular events that link mTOR and sperm methylome. It may take many years and significant effort of many research groups to dissect the whole cascade. It is worth mentioning that understanding of a complete cascade involved in pathology is not needed to develop efficient therapies if the

critical nodes are known. For many common drugs (e.g. metformin) we do not know the full chain of molecular mechanisms but use them successfully.

Thus, we believe that our study is mechanistic as it identified a critical mechanism manipulation of which allows experimental aging and rejuvenation of the sperm methylome. Additionally, it generates new mechanistic questions and hypotheses to be answered in the future.

### **Reviewer #3 (Public Review):**

#### *Summary and Strength:*

*The manuscript by Amir et al. describes that Sertoli-specific inactivation of the mTORC1 and mTORC2 complex by KO of either Raptor or Rictor, respectively, resulted in progressive changes in blood-testis-barrier (BTB) function, testis weight, and sperm parameters, including counts, morphology, mtDNA content and sperm DNA methylation.*

*The described studies are based on the hypothesis that a decline of BTB function with increasing chronological age of a male contributes to the DNA methylation changes that are known to occur in sperm DNA of old males when compared to sperm DNA from isogenic young males. In order to demonstrate the relevance of a functioning BTB for the maintenance of sperm methylation patterns, the authors generated mice with genetically disrupted mTORC2 complex or mTORC1 complex in Sertoli cells and determined sperm methylation patterns in comparison to isogenic wild-type males. In line with previously published scientific literature (e.g. Mok et al., 2013; Dong et al, 2015; and others), the manuscript corroborates that a Sertoli-cell specific deletion of mTORC2 caused a loss of BTB function and a progressive spermatogenic defect. The authors further show that sperm DNA is differentially methylated (DMRs) as a consequence of either a mTORC2 disruption (associated with a loss of BTB function) or following a mTORC1 disruption (BTB function either increased or not leaky) when compared to their isogenic age-matched wt controls. Those DMRs overlap partially with changes in sperm DNA methylation that were found when comparing sperm from 8-week males with sperm isolated from 22-week-old male mice.*

*The authors interpret the observed changes as representative of the sperm DNA methylation changes that occur during normal chronological aging of the male. For an aged control group, the authors use sperm DNA of 22-week-old wild-type males from the mTORC2 and mTORC2 KO breeding and compare the sperm methylation patterns found in sperm from those 22-week males to 8-week young males, that are intended to represent an old and a young cohort, respectively. DNA methylation analysis indicates that a disruption of mTORC2 (& decrease of BTB function) results in increased DNA methylation of sperm DNA, while a disruption of mTORC1 (and proposed increase of BTB tightness, not shown in the manuscript, though) resulted in increased hypomethylation.*

#### *Weaknesses:*

*While the hypothesis and experimental system are interesting and the data demonstrating the relevance of the mTORC2 complex for BTB function is convincing, several open questions limit the evidence that supports the hypothesis that the sperm DNA methylation changes seen in old males are caused by BTB failure following an imbalance of mTOR signaling complexes. The major critique points are the lack of a chronologically old group and the choice of 8 weeks & 22 weeks age of age:*

*- Data illustrating the degree of BTB decline and sperm DNA methylation changes from chronologically "old" male mice is missing. 22-week-old mice are not considered old but are of good and mature breeding age, equivalent to humans in their mid-late twenties.*



*(In the manuscript, the 22-week-old wildtype mice show no evidence of BTB breakdown (Figure 3), so why are their sperm used to represent "aged" sperm?*

*- Adding a group of "old" wild-type mice of 12-14 months of age, which is closer to the end of effective reproduction in mice, more equivalent to 45-59 year-old humans) could be used to illustrate that (a) aging causes a marked decrease in BTB function at this time in mouse life, and that this BTB breakdown chronologically aligns with the age-associated DNA hypermethylation seen in old sperm. Age-matched "old" mTORC1 KO, with a (supposedly) tighter BTB barrier, could then be expected to have a sperm DNA methylation profile closer to that of younger wild-type animals. Such data are currently missing. While the progressive testicular decline observed in the mTORC1 KO (Fig.5) could make it difficult to obtain the appropriately aged mTORC1 KO tissues, it is completely feasible to obtain data from chronologically old wild-type males. (The progressive testicular decline further raises the question of what additional defects the KO causes, and how such additional defects would influence the sperm DNA methylation profile.) The addition of data from an old group to the currently included groups could strengthen the interpretation that the observations in the BTB-defective mTORC2 KO mice are modelling an age-related testicular decline, provided that the DMRs seen in the chronologically old group significantly overlap with the BTB-defective changes.*

*- In the current form, the described differences in sperm DNA methylation are based on comparisons between pubertal mice (8 weeks) and mature but not old adult males (22 weeks), while a chronologically "old" group is missing from the data sets and comparisons. Thus, it appears that the described sperm methylation changes reflect developmental changes associated with normal maturation and not necessarily declining sperm quality due to aging. (Sperm obtained from 8-week-old mice likely were generated, at least in part, during the 1st wave of spermatogenesis, which is known to differ from the continuously proceeding spermatogenesis during the remainder of the mature life. During the 1st wave of spermatogenesis, Sertoli cells are known to undergo gene expression changes which could contribute to varying degrees of BTB function, and thus have effects on the sperm DNA methylation profiles of such 1st wave sperm.)*

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*How do those numbers and profiles compare to the shown samples?*

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*The rebuttal letter and public response indicate the authors' reluctance to consider the limitations of their study, i.e. having chosen chronologically young animals to demonstrate a sperm aging effect and indicate that they are not willing to include adequate controls.*

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*The authors' claim that age-related DNA methylation changes in sperm occur in linear fashion and that the changes are somewhat proportional with chronological age is in*



*stark contrast of the claim that a decline of the BTB in old animals is causative for age-related sperm epigenetic changes, putting the relevance of the whole study in question.*

We are thankful to the reviewer for agreeing to review our revised manuscript. We disagree with the evaluation provided by the reviewer, however.

First, the reviewer misinterpreted the hypothesis of the study, although it is formulated in the last sentence of the Introduction: “... we hypothesized that the balance of mTOR complexes in Sertoli cells may also play a significant role in age-dependent changes in the sperm epigenome.” Instead, the reviewer assigned a different hypothesis to our study (that BTB integrity changes are responsible for age-dependent changes in sperm DNA methylation) and criticized us for not providing clear testing of this hypothesis.

To clarify, we believe that our study provides high-quality testing of OUR hypothesis as we demonstrated experimentally that manipulations of mTOR complexes balance in Sertoli allow acceleration and deceleration of epigenetic aging of sperm. Additionally, our study generated a hypothesis that BTB permeability may mediate the effects of the mTOR pathway on sperm methylome. This second hypothesis is to be tested in the future research.

We also disagree with the reviewer's interpretation of the aging process as an abrupt transition from a young, healthy, and undamaged state to an old, moribund, and damaged state. The whole body of biogerontological knowledge suggests instead steady accumulation of damage over lasting periods of time. For example, this understanding of steady change at the molecular level allowed the development and successful use of epigenetic clock and other molecular clock models, including several variants of sperm epigenetic clocks. These models clearly demonstrate linear or semi-linear accumulation in DNA-methylation changes in various tissues and biological species across the whole lifespan. It is reasonable to assume that BTB permeability decreases with age steadily as well and that in younger animals this decrease may be not easily detected by the existing analytical methods. Experimental data showing the dynamics of the BTB deterioration over age do not exist to our knowledge although it was demonstrated that older animals have loose BTB as compared with young. We agree with the reviewer that future studies testing the role of BTB deterioration for sperm methylome aging will need to provide such evidence. It was not the subject of the current study, however.

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The following is the authors' response to the original reviews.

**Reviewer #1 (Public Review):**

*In the manuscript "Mechanistic target of rapamycin (mTOR) pathway in Sertoli cells regulates age-dependent changes in sperm DNA methylation", the authors proposed to test if the balance of mTOR complexes in Sertoli cells may play a significant role in age-dependent changes in the sperm epigenome. The paper could be of interest and has a good scientific aim but there are too many drawbacks that hamper the initial enthusiasm. All sections need extensive revision. The paper is mostly descriptive without a mechanistic-orientated explanation for the observed results.*

*Specific comments:*

*(1) The abstract is poorly written. There is a lot of unnecessary introduction that does not provide a rationale for the work. It is not possible to understand the experimental approach or the major data just by reading the abstract. It does not clearly represent the work.*

- We have added details of experimental design and results to the abstract and reduced the introductory part of the abstract.

*(2) The introduction is somewhat vague and does not provide a clear rationale for the hypothesis. There should be more focus more on the role of mTOR in Sertoli cells that goes far beyond BTB. That will give more focus on mTOR. Then it is important to focus on BTB and mTOR: what is known? What is the gap and how can it be solved? Several relevant references are missed concerning mTOR and Sertoli cells.*

- The goal of this study was not to explore all potential roles of mTOR pathway in Sertoli cells, but to test if shifts in the balance of mTOR complexes regulate (accelerate/decelerate) epigenetic aging of sperm. As such, we disagree with the reviewer and consider that the current Introduction provides a focused rationale for the study.

*(3) The Material and Methods section needs improvement. There is much important information missing. For instance: how many animals were used per group and how was the breeding done? At what age? Statistical analysis should be explained in detail.*

- The number of animals was clearly stated in the original manuscript. We have added details of breeding and statistical analysis.

*(4) The results description could be improved. It is vague without highlighting how much difference was detected. The results should be numerically described when possible and the differences should be highlighted. A 10% difference may be significant but not biologically relevant. To correctly evaluate the differences it is important to describe them with some degree of detail.*

- For all DNA methylation experiments we provide numerical characteristics of methylation changes, including numbers of DMRs, % change, significance, correlation coefficients. We believe that only age- and genotype-associated changes in reproductive parameters were not characterized in our manuscript in detail. We have added Table 1 to provide these numbers.

*(5) There is no discussion of the data. The authors just summarize their findings without a comprehensive analysis of the literature and how the effects can be mediated. mTOR interacts with different pathways (mTORC1 and mTORC2 are even mediators of distinct pathways). This would be very relevant to discuss. In addition, there are many study limitations not discussed. There is no clear mechanistic explanation of the way by which the mTOR pathway in Sertoli cells regulates age-dependent changes in sperm DNA methylation. The paper seems preliminary.*

- We have added an additional paragraph to the discussion to highlight a potential molecular mechanism that links mTOR pathway with the sperm epigenome.

*(6) Figure 1 is too simple and does not provide any schematic support for the text.*

- We disagree with the reviewer and believe that the figure represents a good visualization of our hypothesis useful for the perception of the study.

*(7) Figure 2 lacks some detail. For instance, how many animals were used for each step?*

- Numbers of animals are provided in the text of the paper.

(8) Taking into consideration the roles of mTOR on sperm, particularly mTORC1, it is not clear whether there were any differences in sperm motility.

- We did not assess sperm motility in this study.

**Reviewer #2 (Public Review):**

*In this study, the authors hypothesized that the balance of mTOR complexes in Sertoli cells may also play a significant role in age-dependent changes in the sperm epigenome. To test this hypothesis, the authors use transgenic mice with manipulated activity of mTOR complexes in Sertoli cells. These results suggest that the mTOR pathway in Sertoli cells may be used as a novel target of therapeutic interventions to rejuvenate the sperm epigenome in advanced-age fathers.*

*The authors attempt to demonstrate that the balance of mTOR complexes in Sertoli cells regulates the rate of sperm epigenetic aging. The authors have effectively met their research objectives, and their conclusions are supported by the data presented.*

- We are very thankful for the positive evaluation of our study.

**Reviewer #3 (Public Review):**

*Summary and Strength:*

*The manuscript by Amir et al. describes that Sertoli-specific inactivation of the mTORC1 and mTORC2 complex by KO of either Raptor or Rictor, respectively, resulted in progressive changes in blood-testis-barrier (BTB) function, testis weight, and sperm parameters, including counts, morphology, mtDNA content and sperm DNA methylation.*

*The described studies are based on the hypothesis that a decline of BTB function with increasing chronological age of a male contributes to the DNA methylation changes that are known to occur in sperm DNA of old males when compared to sperm DNA from isogenic young males. In order to demonstrate the relevance of a functioning BTB for the maintenance of sperm methylation patterns, the authors generated mice with genetically disrupted mTORC2 complex or mTORC1 complex in Sertoli cells and determined sperm methylation patterns in comparison to isogenic wild-type males. In line with previously published scientific literature (e.g. Mok et al., 2013; Dong et al, 2015; and others), the manuscript corroborates that a Sertoli-cell specific deletion of mTORC2 caused a loss of BTB function and a progressive spermatogenic defect. The authors further show that sperm DNA is differentially methylated (DMRs) as a consequence of either a mTORC2 disruption (associated with a loss of BTB function) or following a mTORC1 disruption (BTB function either increased or not leaky) when compared to their isogenic age-matched wt controls. Those DMRs overlap partially with changes in sperm DNA methylation that were found when comparing sperm from 8-week males with sperm isolated from 22-week-old male mice.*

*The authors interpret the observed changes as representative of the sperm DNA methylation changes that occur during normal chronological aging of the male. For an aged control group, the authors use sperm DNA of 22-week-old wild-type males from the mTORC2 and mTORC2 KO breeding and compare the sperm methylation patterns found in sperm from those 22-week males to 8-week young males, that are intended to represent an old and a young cohort, respectively. DNA methylation analysis indicates that a disruption of mTORC2 (& decrease of BTB function) results in increased DNA methylation of sperm DNA, while a disruption of mTORC1 (and proposed increase of BTB tightness, not shown in the manuscript, though) resulted in increased hypomethylation.*

### Weaknesses:

*While the hypothesis and experimental system are interesting and the data demonstrating the relevance of the mTORC2 complex for BTB function is convincing, several open questions limit the evidence that supports the hypothesis that the sperm DNA methylation changes seen in old males are caused by BTB failure following an imbalance of mTOR signaling complexes. The major critique points are the lack of a chronologically old group and the choice of 8 weeks & 22 weeks age of age:*

*- Data illustrating the degree of BTB decline and sperm DNA methylation changes from chronologically "old" male mice is missing. 22-week-old mice are not considered old but are of good and mature breeding age, equivalent to humans in their mid-late twenties. (In the manuscript, the 22-week-old wildtype mice show no evidence of BTB breakdown (Figure 3), so why are their sperm used to represent "aged" sperm?*

*- Adding a group of "old" wild-type mice of 12-14 months of age, which is closer to the end of effective reproduction in mice, more equivalent to 45-59 year-old humans) could be used to illustrate that (a) aging causes a marked decrease in BTB function at this time in mouse life, and that this BTB breakdown chronologically aligns with the age-associated*

*DNA hypermethylation seen in old sperm. Age-matched "old" mTORC1 KO, with a (supposedly) tighter BTB barrier, could then be expected to have a sperm DNA methylation profile closer to that of younger wild-type animals. Such data are currently missing. While the progressive testicular decline observed in the mTORC1 KO (Fig.5) could make it difficult to obtain the appropriately aged mTORC1 KO tissues, it is completely feasible to obtain data from chronologically old wild-type males. (The progressive testicular decline further raises the question of what additional defects the KO causes, and how such additional defects would influence the sperm DNA methylation profile.) The addition of data from an old group to the currently included groups could strengthen the interpretation that the observations in the BTB-defective mTORC2 KO mice are modelling an age-related testicular decline, provided that the DMRs seen in the chronologically old group significantly overlap with the BTB-defective changes.*

*- In the current form, the described differences in sperm DNA methylation are based on comparisons between pubertal mice (8 weeks) and mature but not old adult males (22 weeks), while a chronologically "old" group is missing from the data sets and comparisons. Thus, it appears that the described sperm methylation changes reflect developmental changes associated with normal maturation and not necessarily declining sperm quality due to aging. (Sperm obtained from 8-week-old mice likely were generated, at least in part, during the 1st wave of spermatogenesis, which is known to differ from the continuously proceeding spermatogenesis during the remainder of the mature life. During the 1st wave of spermatogenesis, Sertoli cells are known to undergo gene expression changes which could contribute to varying degrees of BTB function, and thus have effects on the sperm DNA methylation profiles of such 1st wave sperm.)*

*- It is unclear why the aging-related DMRs between the 8 and 22-week-old wild-type mice vary so dramatically between the two wild-type groups derived from the mTORC1 and the mTORC2 breeding (Fig. S4). If the main difference was due to mTORC1 or mTORC2 activity, both wildtype groups should behave very similarly. Changes seen in a truly "old" mouse (e.g. 20 weeks to 56 weeks), changes in "young mTORC1" and in "old mTORC2" are missing. How do those numbers and profiles compare to the shown samples?*

*Some general comments regarding the chosen age of animals:*

- As mentioned, sperm from 8-week-old mice represent many sperm that were produced in the 1st wave of spermatogenesis; 22-week-old mice are not considered chronologically old mice, but mature and "relatively" young animals. 18-24 month-old mice are considered to be equivalent to 56-69 year-old humans, and might be more suitable to detect aging effects. "Old mice" for study purposes should be at least 12-14 months of age, ideally >18 months of age. 22 weeks (5 months of age) are mice at good breeding age, but still considered mature adults, not old males, and therefore are not expected to show typical aging health problems (like declining fertility).

Even the cited reference (Flurkey et al. 2007) defines that "... mice used a reference group for "young mice" should be at least 3 months of age (~ 13 weeks), i.e. fully sexually mature. The authors specifically state: " The young adult group should be at least 3 months old because, although mice are sexually mature by 35 days, relatively rapid maturational growth continues for most biologic processes and structures until about 3 months. The upper age range for the young adult group is typically about 6 months. ... For the middleaged group, 10 months is typically the lower limit.... The upper age limit for the middleaged group is typically 14-15 months, because at this age, most biomarkers still have not changed to their full extent, and some have not yet started changing. For the old group, the lower age limit is 18 months because age-related change for almost all biomarkers of aging can be detected by then. The upper limit is 22-26 months, depending on the genotype." According to this reference, mice up to 6 months of age are generally considered "mature adults" (equivalent to humans 20-30 yrs), mice of 10-14 month are "middle-aged adults" (equivalent to ~38-47 human years) and 18-24 month mice are "old" (equivalent to human of 56-69 yrs.).

Going on these commonly used age ranges, it is unclear why the authors used 8-week-old mice (generally considered pubertal to late adolescent age) as young mice and 5-month-old mice as "old mice".

Differences seen between these cohorts most likely do not reflect aging, but more likely reflect changes associated with normal developmental maturation, since testis and epididymides continue to grow until about 10-11 weeks of age.

- The DMRs identified between 8 and 22-week-old animals could represent DMRs that are dependent on developmental maturation more than being changed in an "age-dependent" manner (in the sense of increased chronological age). This interpretation is congruent with the fact that those DMRs are enriched for developmental categories.

- We are thankful to the reviewer for a detailed explanation of their disagreement with the ages of mice used in this study. In short, the reviewer suggests that our older group (22 weeks) is not old enough to represent aged animals and our young group (8 weeks) may still have spermatozoa from the first wave of spermatogenesis, and as such the observed differences between the 2 ages cannot be considered as aging-related but rather may represent different stages of maturation of the reproductive system. At the first glance this criticism looks valid.

However, to design our experiments we used our data that was not included to this manuscript initially. These data demonstrated that age dependent changes in sperm DNA are linearly or semi linearly associated with age in the age range from 56 to 334 days. Thus, within this interval any 2 ages, distant enough to register the difference in DNA methylation, can be used to assess age dependent changes in DNA methylation and changes in the rates of epigenetic aging of sperm in response to genetic manipulations. We have added these results now, - see "Identification of agedependent patterns in sperm DNA methylation" section in Material and Methods and "Patterns of age-dependent changes in sperm DNA methylation" in Results. We also consider that the reviewer's suggestion that sperm from 8-week-old mice

represents the first wave of spermatogenesis does not have ground. Indeed, C57BL/6 mice first have fertile sperm in cauda epididymis at 37 days of age [1], 19 days earlier than the age of 56 days (8 weeks) at which sperm was collected in our study in the youngest group of mice. Given that young C57BL/6 mice ejaculate spontaneously around 3 times per 5 days [2], 8 weeks old mice have ejaculated > 10 times since the first wave of spermatogenesis before the sperm was collected for our study, making negligibly small the chances of survival of any first wave sperm in their cauda epididymides to the age of 8 weeks. We have added this information to the text.

(1) Mochida, K.; Hasegawa, A.; Ogonuki, N.; Inoue, K.; Ogura, A. Early Production of Offspring by in Vitro Fertilization Using First-Wave Spermatozoa from Prepubertal Male Mice. *J. Reprod. Dev.* 2019, 65, 467–473, doi:10.1262/jrd.2019-042.

(2) Huber, M.H.; Bronson, F.H.; Desjardins, C. Sexual Activity of Aged Male Mice: Correlation with Level of Arousal, Physical Endurance, Pathological Status, and Ejaculatory Capacity. *Biol. Reprod.* 1980, 23, 305–316, doi:10.1095/biolreprod23.2.305.

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